

Tensor partially linear additive quantile regression

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ARTICLE INFO

AMS 2020 subject classifications:

primary 62G05

secondary 62F12

Keywords:

Quantile regression

Tensor regression

Region selection

Asymptotic theory

ABSTRACT

Quantile regression has become indispensable for robust modeling in high-dimensional data, traditionally focusing on vector-valued covariates and scalar-valued responses. However, the recent proliferation of tensor-structured data in fields such as cancer studies, digital marketing, and environmental monitoring, exposes the limitations of these traditional methods. In this paper, we introduce a tensor partially linear additive quantile regression model for tensor-structured data. This model integrates vector-valued and tensor-valued covariates in a unified framework by using nonparametric additive functions to capture the nonlinear effect of the vector-valued covariate and a tensor inner product structure to represent the linear effect of the tensor-valued covariate. We develop an estimation procedure that combines the B-spline approximation with the CANDECOMP/PARAFAC (CP) decomposition and employs a nonconvex penalty to encourage sparsity, thereby selecting the active regions in the tensor-valued covariate. The estimator is computed via an alternating direction method of multipliers (ADMM) algorithm. We establish the rate of convergence and the oracle property of the proposed estimator. Extensive simulations confirm its superior finite-sample performance, and an application to integrated genome analysis further demonstrates its value in practice.

1. Introduction

Quantile regression, pioneered by Koenker and Bassett Jr (1978), has been a robust and widely embraced technique for modeling the conditional distribution of the response variable given covariates. Its popularity stems from its ability to provide a comprehensive analysis of the conditional distribution of the response, model complex heterogeneous relationships, accommodate varying sparsity levels across quantiles, and exhibit robustness to heavy-tailed distributions (Koenker et al., 2017). The versatility of the quantile regression model has made it a formidable tool across various disciplines, including financial risk management (Bayer, 2018), ecological environmental science (Opoku and Aluko, 2021), and medical diagnosis (Hong et al., 2019). However, relationships between covariates and the response variable in real-world scenarios often exhibit nonlinearity, necessitating the development of nonparametric or semiparametric quantile regression models capable of nonlinear modeling. Examples include single-index quantile regression (Wu et al., 2010), additive quantile regression (Horowitz and Lee, 2005), and generalized additive quantile regression (Yee, 2004). While existing research has extensively explored both the statistical properties and practical applications of quantile regression, the majority of studies focus on vector-valued covariates, leaving a gap in addressing tensor-structured data.

Traditional regression methods typically assume covariates as vector-valued and the response variable as scalar-valued. However, with the increasing prevalence of tensor data in modern statistical analysis, including applications such as cancer studies (Chang et al., 2021), network analysis (Hoff, 2015), and spatio-temporal data analysis (Liang et al., 2023), tensor regression has emerged as a prominent approach where covariates or responses are represented in tensor form. Zhou et al. (2013) proposed a tensor regression model that employed a low-rank decomposition, provided asymptotic properties, and introduced the concept of region selection to identify influential active regions on the

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response variable within tensor-valued covariates, akin to variable selection in traditional vector-valued regression. However, the statistical properties of region selection for tensor regression remain underexplored.

The exploration of tensor quantile regression models is still in its early stages. Lu et al. (2020) pioneered the introduction of the quantile tensor regression model, presenting its asymptotic normality and non-asymptotic estimation error bound under the lasso penalty. Subsequently, Ke et al. (2023) introduced a tensor regression model tailored for longitudinal data and provided its asymptotic properties. Liu et al. (2024) developed a novel tensor quantile regression model based on the tensor train decomposition, and further established its consistency, asymptotic normality, and non-asymptotic error bound. Lu et al. (2024) examined the quantile tensor regression model under the convex regularization penalty, analyzing the risk bound of the estimator. These works focus on parametric tensor regression models, but the exploration of nonparametric tensor quantile regression remains nascent. Recently, only Lu (2022) proposed a tensor additive quantile regression model and employed the group lasso to induce sparsity, but did not consider related key problems such as region selection and selection consistency.

In the recently emerging application of integrated genome analysis utilizing tensor-structured data (Fang, 2019; Chang et al., 2021; Chen et al., 2025), tensor-valued covariates typically comprise measurements of thousands of genes across multiple omics, whereas vector-valued covariates contain measurements of clinical or environmental variables, such as age and gender. In the integrated genome analysis example in Section 5, an individual's smoking history exhibits a potentially nonlinear effect and can be modeled nonparametrically. This motivates us to propose a tensor partially linear additive quantile regression (TAQR) model to analyze this integrated genome dataset. This model incorporates vector-valued covariates and tensor-valued covariates within a unified framework, where nonlinear effects of vector-valued covariates are captured by nonparametric additive functions, while linear effects of tensor-valued covariates are represented via a tensor inner product structure. We employ B-spline approximations for the nonparametric component and impose a CP decomposition on the tensor-valued covariates to reduce dimensionality, and propose a computationally efficient and statistically consistent estimation procedure. Furthermore, we adopt a nonconvex penalty to achieve sparsity, thereby selecting active regions within the tensor-valued covariates. Additionally, we design an efficient ADMM-based algorithm for solving the proposed estimator. Theoretically, we establish the asymptotic properties of the proposed estimator, including the rate of convergence and the oracle property, thereby ensuring the selection consistency of the proposed estimator.

The remainder of this paper is organized as follows. In Section 2, we introduce some notations and preliminaries, present the proposed TAQR model and estimator, establish the asymptotic properties of the estimator, construct the confidence interval, and provide an efficient algorithm for solving the estimator. In Section 3, we conduct numerical simulations to evaluate the finite-sample performance. In Section 4, we illustrate the method via a case study on a genomic dataset. Lastly, Section 5 provides concluding remarks and outlines potential future avenues of research. The appendix provides the proofs of theorems.

2. Methodology

2.1. Notations and Preliminaries

Tensors, the multidimensional generalization of matrices, provide a natural framework for representing high-dimensional data. Throughout this paper, we follow the tensor notations and conventions established by Kolda and Bader (2009), and Liu et al. (2022).

We adopt the following notation: tensors are denoted by boldface Euler script letters such as $\mathcal{X}, \mathcal{Y}$; matrices by boldface capital letters such as $\mathbf{X}, \mathbf{Y}$; vectors by boldface lowercase letters such as $\mathbf{x}, \mathbf{y}$. An M -order tensor $\mathcal{X} \in \mathbb{R}^{p_1 \times \dots \times p_M}$ has M modes, with p_m representing the dimension of the m -th mode. The $(i_1, \dots, i_M)$ -th element of $\mathcal{X}$ is denoted as $\mathcal{X}_{i_1 \dots i_M}$. We introduce several fundamental operations for M -order tensors:

1. Vectorization: The vectorization of a tensor $\mathcal{A} \in \mathbb{R}^{p_1 \times \dots \times p_M}$ is denoted as $\text{VEC}(\mathcal{A}) \in \mathbb{R}^{\prod_{m=1}^M p_m}$, where the $(j_1, \dots, j_M)$ -th element of $\mathcal{A}$ is mapped to the j -th element of $\text{VEC}(\mathcal{A})$, and $j = 1 + \sum_{m=1}^M [(j_m - 1) \prod_{m'=1}^{m-1} p_{m'}]$.
2. Mode- m unfolding: The mode- m unfolding of a tensor $\mathcal{A} \in \mathbb{R}^{p_1 \times \dots \times p_M}$ is denoted as $\mathcal{A}_{(m)} \in \mathbb{R}^{p_m \times (p_1 \dots p_{m-1} p_{m+1} \dots p_M)}$, where the $(j_1, \dots, j_M)$ -th element of $\mathcal{A}$ is mapped to the (j_m, j) -th element of $\mathcal{A}_{(m)}$, and $j = 1 + [\sum_{k=1}^M (j_k - 1) \prod_{k'=1(\neq m)}^{k-1} p_{k'}]$.
3. Tensor inner product: For two tensors $\mathcal{A}, \mathcal{B} \in \mathbb{R}^{p_1 \times \dots \times p_M}$, their inner product is $\langle \mathcal{A}, \mathcal{B} \rangle = \text{VEC}(\mathcal{A})^\top \text{VEC}(\mathcal{B})$.
4. Tensor ℓ_1 norm: For a tensor $\mathcal{A} \in \mathbb{R}^{p_1 \times \dots \times p_M}$, its ℓ_1 norm is defined as $\|\mathcal{A}\|_1 = \sum_{i_1=1}^{p_1} \dots \sum_{i_M=1}^{p_M} |\mathcal{A}_{i_1 \dots i_M}|$.

5. Tensor Frobenius norm: For a tensor $\mathcal{A} \in \mathbb{R}^{p_1 \times \dots \times p_M}$, its Frobenius norm is defined as $\|\mathcal{A}\|_F = \langle \mathcal{A}, \mathcal{A} \rangle^{1/2}$.
6. CP decomposition: The CP decomposition decomposes a tensor $\mathcal{A}$ into a sum of several rank-1 tensors: $\mathcal{A} = \sum_{r=1}^R \alpha_{1,r} \circ \dots \circ \alpha_{M,r}$, where $\alpha_{m,r} \in \mathbb{R}^{p_m}$ for $m = 1, \dots, M$, $\circ$ denotes the vector outer product, and R is called the CP rank. For notational convenience, we represent the CP decomposition compactly as $\mathcal{A} = [\![\mathbf{A}_1, \dots, \mathbf{A}_M]\!]$, where each factor $\mathbf{A}_m = (\alpha_{m,1}, \dots, \alpha_{m,R})$ is a $p_m \times R$ matrix.

Lemma 1 (Kolda (2006)). *If a tensor $\mathcal{A}$ admits the CP decomposition, then*

$$\mathcal{A}_{(m)} = \mathbf{A}_m (\mathbf{A}_M \odot \dots \odot \mathbf{A}_{m+1} \odot \mathbf{A}_{m-1} \odot \dots \odot \mathbf{A}_1)^\top,$$

where $\odot$ denotes the matrix Khatri-Rao product.

2.2. Tensor Partially Linear Additive Quantile Regression Model

In this section, we introduce a flexible tensor partially linear additive quantile regression model for modeling the relationship between scalar-valued response $y \in \mathbb{R}$, vector-valued covariate $\mathbf{z} = (z_1, \dots, z_{p_0})^\top \in \mathbb{R}^{p_0}$, and tensor-valued covariate $\mathcal{X} \in \mathbb{R}^{p_1 \times \dots \times p_M}$. Assume that $F(\cdot | \mathbf{z}, \mathcal{X})$ is the conditional distribution function of response y given covariates $\mathbf{z}$ and $\mathcal{X}$. Then the τ -th conditional quantile, for $\tau \in (0, 1)$, of response y , given covariates $\mathbf{z}$ and $\mathcal{X}$, is $Q_\tau(y | \mathbf{z}, \mathcal{X}) = \inf \{t : F(t | \mathbf{z}, \mathcal{X}) \geq \tau\}$. We propose to model the conditional quantile by

$$Q_\tau(y | \mathbf{z}, \mathcal{X}) = g(\mathbf{z}) + \langle \mathcal{B}, \mathcal{X} \rangle, \quad (1)$$

where $g(\mathbf{z}) = \alpha + \sum_{j_0=1}^{p_0} g_{j_0}(z_{j_0})$ with $\alpha \in \mathbb{R}$, and $\mathcal{B}$ is the coefficient tensor of compatible dimensions. It is assumed that $g_{j_0}, j_0 = 1, \dots, p_0$, satisfy $\mathbb{E}[g_{j_0}(z_{j_0})] = 0$ for identification. Since the intercept term can be easily estimated by the global quantile of the response variable and is highly tractable both theoretically and computationally, we omit it here for simplicity of notation.

Given an independent and identically distributed sample $\{y_i, \mathbf{z}_i, \mathcal{X}_i\}_{i=1}^n$ of $(y, \mathbf{z}, \mathcal{X})$, let $\varepsilon_i = y_i - g^*(\mathbf{z}_i) - \langle \mathcal{B}^*, \mathcal{X}_i \rangle$, then ε_i satisfies $P(\varepsilon_i \leq 0 | \mathbf{z}_i, \mathcal{X}_i) = \tau$, and $y_i = g^*(\mathbf{z}_i) + \langle \mathcal{B}^*, \mathcal{X}_i \rangle + \varepsilon_i$, where $g^*(\cdot)$ and $\mathcal{B}^*$ are the true nonlinear additive function and true coefficient tensor, respectively.

In model (1), the coefficient tensor contains $p_n = \prod_{m=1}^M p_m$ parameters, which typically far exceed the sample size. In the high dimensional scenario, an effective method is to impose a low-rank assumption on the coefficient tensor, and then use the idea of dimensionality reduction to reduce the number of parameters, thereby improving computational efficiency (Lu et al., 2020; Li and Zhang, 2021; Hao et al., 2021; Liu et al., 2024; Zhou et al., 2024). To achieve parameter dimension reduction while preserving the tensor intrinsic structure, we impose a low-rank assumption on the coefficient tensor and employ the CP decomposition. This leads to the following semiparametric quantile regression model with tensor decomposition,

$$Q_\tau(y | \mathbf{z}, \mathcal{X}) = g(\mathbf{z}) + \langle [\![\mathbf{B}_1, \dots, \mathbf{B}_M]\!] , \mathcal{X} \rangle, \quad (2)$$

where $\mathbf{B}_m \in \mathbb{R}^{p_m \times R}$ for $m = 1, \dots, M$. As a result of decomposition, the number of parameters in model (2) reduces to $R \sum_{m=1}^M p_m$, which is substantially smaller than $p_n = \prod_{m=1}^M p_m$.

2.3. Estimation

In order to estimate the model (2), we first use a linear combination of B-spline basis functions to approximate the unknown nonlinear additive function $g(\cdot)$, and then adopt a nonconvex penalty function on the coefficient tensor $[\![\mathbf{B}_1, \dots, \mathbf{B}_M]\!]$ to simultaneously perform estimation and region selection.

Let $\boldsymbol{\pi}(t) = [b_1(t), \dots, b_{k_n+l+1}(t)]^\top$ be a vector of normalized B-spline basis functions of order $l+1$ with k_n quasi-uniform internal knots on $[0, 1]$. Then the unknown function $g_{j_0}(\cdot)$ can be approximated by $g_{j_0}(z_{j_0}) \approx \boldsymbol{\pi}(z_{j_0})^\top \boldsymbol{\xi}_{j_0}$, where $\boldsymbol{\xi}_{j_0} \in \mathbb{R}^{J_n}$ for $j_0 = 1, \dots, p_0$, and $J_n = k_n + l + 1$. Define $\boldsymbol{\Pi}(\mathbf{z}) = [1, \boldsymbol{\pi}(z_1)^\top, \dots, \boldsymbol{\pi}(z_{p_0})^\top]^\top \in \mathbb{R}^{p_0 J_n + 1}$ as the B-spline vector for all vector-valued covariates.

To estimate the spline coefficient $\boldsymbol{\xi} = (\xi_0, \xi_1^\top, \dots, \xi_{p_0}^\top)^\top$ and the coefficient tensor $\mathcal{B} = [\![\mathbf{B}_1, \dots, \mathbf{B}_M]\!]$, we adopt the check loss function, $\rho_\tau(u) = u[\tau - I(u < 0)]$, proposed by Koenker and Bassett Jr (1978), where $I(\cdot)$ is the indicator function. Then, the objective function is defined as

$$L(\boldsymbol{\xi}, [\![\mathbf{B}_1, \dots, \mathbf{B}_M]\!]) = n^{-1} \sum_{i=1}^n \rho_\tau[y_i - \boldsymbol{\Pi}(\mathbf{z}_i)^\top \boldsymbol{\xi} - \langle [\![\mathbf{B}_1, \dots, \mathbf{B}_M]\!] , \mathcal{X}_i \rangle]. \quad (3)$$

Define $\mathcal{H}_r$ as the collection of functions satisfying the Hölder condition. That is, for any $h(\cdot) \in \mathcal{H}_r$, there exists positive constant C such that $|h^{(\kappa)}(x_1) - h^{(\kappa)}(x_2)| \leq C |x_1 - x_2|^v$, where $0 \leq x_1, x_2 \leq 1$, κ is a positive integer, $v \in (0, 1]$, and $r = \kappa + v$. Assume that for $r \geq 2$, the nonparametric components $g_{j_0} \in \mathcal{H}_r$ for $j_0 = 1, \dots, p_0$. According to Schumaker (1981), there exists $\xi^* \in \mathbb{R}^{p_0 J_n + 1}$ such that

$$\sup_{z_i} \left| \Pi(z_i)^\top \xi^* - g(z_i) \right| = O(k_n^{-r}).$$

To encourage the sparsity of the coefficient tensor and select the active region in the tensor-valued covariate, we propose the following element-wise penalized TAQR estimator

$$(\hat{\xi}, \llbracket \hat{\mathbf{B}}_1, \dots, \hat{\mathbf{B}}_M \rrbracket) = \arg \min_{\xi, \mathbf{B}_1, \dots, \mathbf{B}_M} L(\xi, \llbracket \mathbf{B}_1, \dots, \mathbf{B}_M \rrbracket) + \sum_{j_1=1}^{p_1} \dots \sum_{j_M=1}^{p_M} P_\lambda(|\llbracket \mathbf{B}_1, \dots, \mathbf{B}_M \rrbracket_{j_1, \dots, j_M}|), \quad (4)$$

where $P_\lambda(\cdot)$ is the penalty function with tuning parameter λ .

Remark 1. Compared with the mode-wise penalty approaches (Lu et al., 2020; Li and Zhang, 2021; Liu et al., 2024), the proposed estimator (4) imposes the penalty function directly on the coefficient tensor in an element-wise manner, rather than on its decomposition factors. The proposed element-wise penalized estimator has three advantages. First, element-wise penalization on $\mathbf{B}$ provides a more faithful identification of the truly active regions and avoids the potential reconstruction error introduced by mapping penalized factors back to the coefficient tensor. Second, the theoretical analysis is simplified, since sparsity is characterized at the level of the target coefficient tensor $\mathbf{B}$. Third, although the penalty is imposed element-wise on the coefficient tensor $\mathbf{B}$, the proposed method remains computational advantage relative to the traditional vector-valued regression approach. This is because the coefficient tensor is estimated at the mode-wise level, and the element-wise formulation is used only in the penalty term to achieve better region selection performance. We demonstrate the computational advantage through numerical simulation in Section 3.6.

The choice of regularization penalty plays a pivotal role in penalized estimation. Among various regularization methods, the lasso (least absolute shrinkage and selection operator, Tibshirani (1996)), or ℓ_1 norm penalty, has emerged as a predominant technique in high-dimensional regression analysis. However, lasso suffers from estimation bias and tends to over-penalize large coefficients. By contrast, appropriately chosen nonconvex penalty functions can effectively mitigate these issues. To obtain the oracle property that guarantees selection consistency, we adopt a general class of nonconvex penalties. Two widely used nonconvex penalty functions are the smoothly clipped absolute deviation penalty (SCAD, Fan and Li (2001)) and the minimax concave penalty (MCP, Zhang (2010)). The SCAD penalty is defined as

$$P_\lambda(x) = \lambda |x| I(0 \leq |x| < \lambda) + \frac{a\lambda |x| - (x^2 + \lambda^2)/2}{a-1} I(\lambda \leq |x| \leq a\lambda) + \frac{(a+1)\lambda^2}{2} I(|x| > a\lambda),$$

for some $a > 2$, and the MCP penalty is given by

$$P_\lambda(x) = \lambda \left(|x| - \frac{x^2}{2a\lambda} \right) I(0 \leq |x| < a\lambda) + \frac{a\lambda^2}{2} I(|x| \geq a\lambda),$$

for some $a > 1$, where the penalty parameter $\lambda > 0$ controls the degree of sparsity, vanishing as the sample size $n \rightarrow \infty$, while the constant a determines how rapidly $P'_\lambda(x)$ decays to zero beyond the threshold $a\lambda$ and is held fixed in the asymptotic analysis. We will employ the SCAD and MCP penalties, and the resulting estimators are coined as TAQR-SCAD and TAQR-MCP, respectively.

2.4. Asymptotic Properties

For statistical inference, we consider the scenario in which the number of components in the tensor-valued covariate, $p_n = \prod_{m=1}^M p_m$, diverges. In the region selection context, while the tensor-valued covariate contains p_n components, only q_n of these components are needed to model the conditional quantile. In other words, the true coefficient tensor $\mathbf{B}^*$ is sparse, with a substantial proportion of its components being zero. Let

$$\mathbb{A} = \left\{ j : j = 1 + \sum_{m=1}^M \left[(j_m - 1) \prod_{k=1}^{m-1} p_k \right], \mathbf{B}_{j_1, \dots, j_M}^* \neq 0, 1 \leq j_m \leq p_m, 1 \leq m \leq M \right\}$$

be the index set of nonzero components and $q_n = |\mathbb{A}|$ be the cardinality of $\mathbb{A}$. The complement of $\mathbb{A}$ is denoted by $\mathbb{A}^c$. An important task is to estimate the index set $\mathbb{A}$.

Before establishing the asymptotic properties of the estimator, we first present some notations and their definitions that will be used throughout our theoretical analysis. Let

$$\boldsymbol{\beta} = [\text{VEC}(\mathbf{B}_1)^\top, \dots, \text{VEC}(\mathbf{B}_M)^\top]^\top, \text{ and} \\ \boldsymbol{\phi} = \boldsymbol{\phi}(\boldsymbol{\beta}) = \text{VEC}(\mathbf{B}) = \text{VEC}(\llbracket \mathbf{B}_1, \dots, \mathbf{B}_M \rrbracket).$$

Without loss of generality, we partition the vectorized coefficient tensor $\boldsymbol{\phi}$ into $\boldsymbol{\phi} = (\boldsymbol{\phi}_{\mathbb{A}}^\top, \boldsymbol{\phi}_{\mathbb{A}^c}^\top)^\top$, where $\boldsymbol{\phi}_{\mathbb{A}}$ represents the sub-vector associated with the active region indexed by $\mathbb{A}$, and $\boldsymbol{\phi}_{\mathbb{A}^c}$ represents the sub-vector associated with the inactive region indexed by $\mathbb{A}^c$. We assume the true parameter $\boldsymbol{\phi}^*$ satisfies $\boldsymbol{\phi}_{\mathbb{A}^c}^* = 0$, and all components of $\boldsymbol{\phi}_{\mathbb{A}}^*$ do not vanish. Let $\tilde{\mathbf{X}} = (\text{VEC}(\mathcal{X}_1), \dots, \text{VEC}(\mathcal{X}_n))^\top \in \mathbb{R}^{n \times p_n}$, and $\tilde{\mathbf{X}}_{\mathbb{A}} \in \mathbb{R}^{n \times q_n}$ be the sub-matrix of $\tilde{\mathbf{X}}$.

We define the following function space

$$\mathcal{G} = \left\{ g(\mathbf{z}) : g(\mathbf{z}) = \alpha + \sum_{j_0=1}^{p_0} g_{j_0}(\mathbf{z}_{j_0}), g_{j_0} \in \mathcal{H}_r, \mathbb{E}[g_{j_0}(\mathbf{z}_{j_0})] = 0, \mathbb{E}[g_{j_0}^2(\mathbf{z}_{j_0})] < \infty \right\}.$$

For $1 \leq i \leq n, 1 \leq j \leq q_n$, let $h_j^*(\cdot) = \arg \inf_{h_j(\cdot) \in \mathcal{G}} \sum_{i=1}^n \mathbb{E}[(\tilde{\mathbf{X}}_{\mathbb{A}ij} - h_j(\mathbf{z}_i))^2]$. Let $\delta_{ij} = \tilde{\mathbf{X}}_{\mathbb{A}ij} - h_j^*(\mathbf{z}_i)$, $\boldsymbol{\delta}_i = (\delta_{i1}, \dots, \delta_{iq_n})^\top \in \mathbb{R}^{q_n}$, $\boldsymbol{\Delta}_n = (\boldsymbol{\delta}_1, \dots, \boldsymbol{\delta}_n)^\top \in \mathbb{R}^{n \times q_n}$. Let $\mathbf{H}$ be the $n \times q_n$ matrix with the (i, j) -th element $H_{ij} = h_j^*(\mathbf{z}_i)$, then $\tilde{\mathbf{X}}_{\mathbb{A}} = \mathbf{H} + \boldsymbol{\Delta}_n$. Let

$$\boldsymbol{\Omega}_\lambda = \text{diag}\{P_\lambda''(|\phi_1^*|), \dots, P_\lambda''(|\phi_{q_n}^*|)\}, \text{ and} \\ \mathbf{c}_n = \{P_\lambda'(|\phi_1^*|)\text{sgn}(\phi_1^*), \dots, P_\lambda'(|\phi_{q_n}^*|)\text{sgn}(\phi_{q_n}^*)\}^\top.$$

The following mild conditions will be used to establish the asymptotic properties.

Condition 1. Given $\mathbf{z}_i, \mathcal{X}_i$, the random error ε_i has the conditional distribution function $F_\varepsilon(\cdot|\mathbf{z}_i, \mathcal{X}_i)$ and the conditional density function $f_\varepsilon(\cdot|\mathbf{z}_i, \mathcal{X}_i)$. $f_\varepsilon(\cdot|\mathbf{z}_i, \mathcal{X}_i)$ is in bounded away from 0 and ∞ uniformly. The derivative function of $f_\varepsilon(\cdot|\mathbf{z}_i, \mathcal{X}_i)$ is also uniformly bounded.

Condition 2. There exist positive constants M_1 and M_2 such that $|\tilde{\mathbf{X}}_{ij}| \leq M_1$ for any $1 \leq i \leq n, 1 \leq j \leq p_n$, and $\mathbb{E}(\delta_{ij}^4) \leq M_2$ for any $1 \leq i \leq n, 1 \leq j \leq q_n$. There exist positive constants C_1 and C_2 such that $C_1 \leq \lambda_{\min}(n^{-1} \tilde{\mathbf{X}}_{\mathbb{A}}^\top \tilde{\mathbf{X}}_{\mathbb{A}}) \leq \lambda_{\max}(n^{-1} \tilde{\mathbf{X}}_{\mathbb{A}}^\top \tilde{\mathbf{X}}_{\mathbb{A}}) \leq C_2, C_1 \leq \lambda_{\min}(n^{-1} \tilde{\mathbf{X}}_{\mathbb{A}}^\top \boldsymbol{\Delta}_n) \leq \lambda_{\max}(n^{-1} \tilde{\mathbf{X}}_{\mathbb{A}}^\top \boldsymbol{\Delta}_n) \leq C_2$.

Condition 3. The B-spline basis satisfies $k_n = O(n^{1/(2r+1)})$.

Condition 4. The parameter space for $\boldsymbol{\phi}$ is bounded.

Condition 5. The true parameter $\boldsymbol{\phi}^*$ satisfies $\min_{1 \leq j \leq q_n} |\phi_j^*|/\lambda \rightarrow \infty$, as $n \rightarrow \infty$.

Condition 6. $a_n = O(n^{-1/2})$ and $b_n \rightarrow 0$ as $n \rightarrow \infty$, where

$$a_n = \max_{1 \leq j \leq p_n} \{P_\lambda'(|\phi_j^*|), \phi_j^* \neq 0\}, \quad b_n = \max_{1 \leq j \leq p_n} \{P_\lambda''(|\phi_j^*|), \phi_j^* \neq 0\}.$$

Condition 7. There exist constants C and D such that $|P_\lambda''(\theta_1) - P_\lambda''(\theta_2)| \leq D|\theta_1 - \theta_2|$, where $\theta_1, \theta_2 > C\lambda$. Meanwhile, $\liminf_{n \rightarrow +\infty} \liminf_{\theta \rightarrow 0+} P_\lambda'(\theta)/\lambda > 0$.

Remark 2. Condition 1 is a classical condition commonly employed in establishing asymptotic properties for quantile regression (Lu et al., 2020; Zhang et al., 2020; Liu et al., 2024; Liang et al., 2025). This condition does not impose any moment assumptions on the error distribution, thereby ensuring the robustness of the estimator against heavy-tailed distributions or outliers. Condition 2 specifies assumptions regarding covariates and the design matrix, which are frequently adopted when examining asymptotic properties of partially linear models (Sherwood and Wang, 2016a; Wang et al., 2023). Condition 3 determines the required rate k_n that achieves the optimal convergence rate for

the nonparametric component, a condition typically used when employing B-splines for nonparametric regression estimation (Lian, 2012; Sherwood and Wang, 2016a; Sherwood and Maidman, 2022). Condition 4 specifies the boundedness assumption on the parameter space, which is typically necessary for quantile tensor regression theory (Lu et al., 2020; Liu et al., 2024). Condition 5 is a minimal signal condition for establishing the oracle property (Fan and Peng, 2004; Jiang et al., 2012). Conditions 6-7 are the regularity conditions on the penalty function, which have been used for regression models with a diverging number of parameters (Fan and Peng, 2004; Jiang et al., 2012; Wang et al., 2014).

Theorem 1. *If Conditions 1 - 7 hold and $p_n^3/n \rightarrow 0$ as $n \rightarrow \infty$, then there is a local minimizer $(\hat{\xi}, \hat{\phi})$ such that*

$$\|\hat{\xi} - \xi^*\|_2 + \|\hat{\phi} - \phi^*\|_2 = O_P\left(\sqrt{\frac{R \sum_{m=1}^M p_m + k_n}{n}}\right).$$

Remark 3. According to Theorem 1, the TAQR estimator achieves a faster rate of convergence compared to the $O_P(\sqrt{(p_n + k_n)/n})$ rate of vector-based quantile regression. This improvement is attributed to the implementation of the tensor decomposition technology. Meanwhile, the divergence of the dimensionality in the linear tensor component affects the convergence rate of $g(\cdot)$. When the dimensionality of the linear tensor component remains fixed, $g(\cdot)$ can attain the optimal rate of convergence $O_P(n^{-2r/(2r+1)})$.

Lemma 2. *Let $\mathbf{J} \in \mathbb{R}^{(\prod_{m=1}^M p_m) \times (R \sum_{m=1}^M p_m)}$ be the Jacobian matrix of $\phi(\beta)$, where*

$$\mathbf{J} = \partial\phi/\partial\beta = [\Pi_1[(\mathbf{B}_M \odot \cdots \odot \mathbf{B}_2) \otimes \mathbf{I}_{p_1}], \Pi_2[(\mathbf{B}_M \odot \cdots \odot \mathbf{B}_3 \odot \mathbf{B}_1) \otimes \mathbf{I}_{p_2}], \dots, \Pi_M[(\mathbf{B}_{M-1} \odot \cdots \odot \mathbf{B}_1) \otimes \mathbf{I}_{p_M}]],$$

and $\Pi_m \in \mathbb{R}^{(\prod_{m=1}^M p_m) \times (\prod_{m=1}^M p_m)}$ is the permutation matrix. We denote the first q_n row sub-matrix of $\mathbf{J}$ as $\mathbf{J}_{\mathbb{A}}$.

Theorem 2. *If Conditions 1 - 7 hold. If $\lambda \rightarrow 0$, $\lambda[n(R \sum_{m=1}^M p_m)^{-1}]^{1/2} \rightarrow \infty$, and $p_n^3/n \rightarrow 0$ as $n \rightarrow \infty$, then with probability tending to one, the local minimizer $(\hat{\phi}_{\mathbb{A}}^\top, \hat{\phi}_{\mathbb{A}^c}^\top)^\top$ in Theorem 1 satisfies that*

- (i) Sparsity: $\hat{\phi}_{\mathbb{A}^c} = 0$;
- (ii) Asymptotic normality:

$$\sqrt{ne_n^\top} \mathbf{K}^{-1/2} (\mathbf{K} + \boldsymbol{\Omega}_\lambda) [\hat{\phi}_{\mathbb{A}} - \phi_{\mathbb{A}}^* + (\mathbf{K} + \boldsymbol{\Omega}_\lambda)^{-1} \mathbf{c}_n] \xrightarrow{d} N(\mathbf{0}, \tau(1 - \tau)),$$

where $\mathbf{e}_n$ is a unit vector, $\mathbf{K} = (\mathbf{P}\mathbf{\Gamma}\mathbf{P}^\top)^{-1}$, $\mathbf{\Gamma} = \mathbf{G}_n^{-1} \mathbf{S}_n \mathbf{G}_n^{-1}$, $\mathbf{G}_n = n^{-1} \mathbf{\Delta}_n^\top \mathbf{D}_n \mathbf{\Delta}_n$, $\mathbf{S}_n = n^{-1} \mathbf{\Delta}_n^\top \mathbf{\Delta}_n$, $\mathbf{D}_n = \text{diag}\{f_\epsilon(0|\mathbf{z}_1, \mathcal{X}_1), \dots, f_\epsilon(0|\mathbf{z}_n, \mathcal{X}_n)\}$, $\mathbf{P} = \mathbf{J}_{\mathbb{A}} (\mathbf{J}_{\mathbb{A}}^\top \mathbf{G}_n \mathbf{J}_{\mathbb{A}})^\dagger \mathbf{J}_{\mathbb{A}}^\top \mathbf{G}_n$. Here $(\cdot)^\dagger$ is the pseudo-inverse.

Remark 4. Theorem 2 demonstrates that the proposed nonconvex penalized TAQR estimator possesses the oracle property, specifically, it correctly identifies the active regions within the tensor-valued covariate. When n is large enough, $\boldsymbol{\Omega}_\lambda = \mathbf{0}$ and $\mathbf{c}_n = \mathbf{0}$ for nonconvex penalty functions. Therefore, the asymptotic normality in Theorem 2 becomes

$$\sqrt{ne_n^\top} (\mathbf{P}\mathbf{\Gamma}\mathbf{P}^\top)^{-1/2} (\hat{\phi}_{\mathbb{A}} - \phi_{\mathbb{A}}^*) \xrightarrow{d} N(\mathbf{0}, \tau(1 - \tau)),$$

which implies that the nonconvex penalized TAQR estimator is as efficient as the oracle estimator.

Remark 5. There are two degenerate cases: (1) When the number of modes of the tensor-valued covariate reduces to $M = 1$, the tensor-valued covariate degenerates to the vector-valued covariate, and our model simplifies to the partially linear additive quantile regression model studied in Sherwood and Wang (2016a). Under the same conditions, the convergence rate in Theorem 1 becomes $O_P(\sqrt{(p_1 + k_n)/n})$, which matches their result, and the over-parameterized structure in Theorem 2 vanishes. (2) When the nonparametric component $g(\cdot)$ is absent, our method reduces to the quantile tensor regression model studied in (Lu et al., 2020). However, our estimator differs from the estimator of (Lu et al., 2020), and consequently the theoretical results also exhibit certain differences.

2.5. Confidence Interval

In Theorem 2, we establish the asymptotic behavior of the proposed TAQR estimator. This asymptotic properties proves highly valuable for conducting statistical inference. Accordingly, we proceed to construct confidence interval for the active region. As in Fan and Li (2001); Fan and Peng (2004), we can obtain the estimation of the covariance matrix of $\hat{\phi}_{\mathbb{A}}$ by the sandwich formula

$$\widehat{\text{cov}}(\hat{\phi}_{\mathbb{A}}) = \frac{1}{n} \mathbf{J}_{\mathbb{A}} \Sigma_1^{-1} \Sigma_2 \Sigma_1^{-1} \mathbf{J}_{\mathbb{A}}^{\top}$$

where $\Sigma_1 = \frac{f_{\varepsilon}(0|\mathbf{z}, \mathbf{X})}{n} \mathbf{J}_{\mathbb{A}}^{\top} \mathbf{X}_{\mathbb{A}}^{\top} \mathbf{X}_{\mathbb{A}} \mathbf{J}_{\mathbb{A}}$, $\Sigma_2 = \frac{1}{n} \sum_{i=1}^n (\tau - I(\hat{\varepsilon}_i < 0))^2 \mathbf{X}_{\mathbb{A},i} \mathbf{X}_{\mathbb{A},i}^{\top}$, and $\hat{\varepsilon}_i = y_i - \Pi(\mathbf{z}_i)^{\top} \hat{\xi} - \langle \hat{\mathbf{B}}, \mathcal{X}_i \rangle$. The $f_{\varepsilon}(0|\mathbf{z}, \mathbf{X})$ can be estimated by the kernel density estimation.

We can conduct the confidence interval at a confidence level $1 - \alpha$ for a component $\phi_{\mathbb{A},j}$ of the active region

$$\text{CI}_j = \left[\hat{\phi}_{\mathbb{A},j} - z_{1-\alpha/2} \sqrt{\widehat{\text{cov}}(\hat{\phi}_{\mathbb{A}})_{jj}}, \hat{\phi}_{\mathbb{A},j} + z_{1-\alpha/2} \sqrt{\widehat{\text{cov}}(\hat{\phi}_{\mathbb{A}})_{jj}} \right]$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ quantile of standard normal distribution.

2.6. An Efficient Algorithm

Solving the estimator (4) presents significant challenges. Firstly, conventional algorithms that decompose tensor regression into iterative mode-wise updates via vector-valued regression (Lu et al., 2020; Li and Zhang, 2021; Liu et al., 2024) are no longer applicable, as the penalty term in estimator (4) does not adhere to mode-wise penalization, thereby decoupling it from the mode-wise updates of the check loss function. Secondly, solving the estimator (4) encounters additional difficulties stemming from the non-smoothness of the check loss function and the non-convexity of the penalty function. To address these challenges, we propose an alternating optimization algorithm based on the ADMM framework (Boyd et al., 2011). Specifically, we employ an alternating optimization approach to iteratively update the spline coefficient ξ and the coefficient tensor $\mathbf{B}$, where the coefficient tensor optimization subproblem is solved within the ADMM framework.

Firstly, the coefficient tensor is updated by

$$\hat{\mathbf{B}}^{(t)} = [\hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)}] = \arg \min_{\mathbf{B}_1, \dots, \mathbf{B}_M} n^{-1} \sum_{i=1}^n \rho_{\tau}(\tilde{y}_i - \langle [\mathbf{B}_1, \dots, \mathbf{B}_M], \mathcal{X}_i \rangle) + \sum_{j_1=1}^{p_1} \dots \sum_{j_M=1}^{p_M} P_{\lambda}(|[\mathbf{B}_1, \dots, \mathbf{B}_M]_{j_1, \dots, j_M}|), \quad (5)$$

where $\tilde{y}_i = y_i - \Pi(\mathbf{z}_i)^{\top} \hat{\xi}^{(t-1)}$ for $i = 1, \dots, n$, and $t > 0$ denotes the t -th iteration.

To solve the coefficient tensor optimization subproblem, we employ the ADMM algorithm. To address the nonsmoothness of the check loss function and the nonconvex penalty function, we introduce two auxiliary variables, $\boldsymbol{\eta}$ and $\mathcal{W}$. It can be shown that formulation (5) is equivalent to the following constrained optimization problem

$$\begin{aligned} \arg \min_{\boldsymbol{\eta}, \mathbf{B}_1, \dots, \mathbf{B}_M, \mathcal{W}} \quad & \mathbb{Q}_{\tau}(\boldsymbol{\eta}) + \sum_{j_1, \dots, j_M} P_{\lambda}(|\mathcal{W}_{j_1, \dots, j_M}|), \\ \text{s.t.} \quad & \boldsymbol{\eta} = \tilde{\mathbf{y}} - \langle \mathcal{X}_n, [\mathbf{B}_1, \dots, \mathbf{B}_M] \rangle_M, \mathcal{W} = \mathbf{B} = [\mathbf{B}_1, \dots, \mathbf{B}_M], \end{aligned} \quad (6)$$

where $\mathbb{Q}_{\tau}(\boldsymbol{\eta}) = n^{-1} \sum_{i=1}^n \rho_{\tau}(\eta_i)$, $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^{\top}$, $\mathcal{X}_n \in \mathbb{R}^{n \times p_1 \times \dots \times p_M}$, and $\langle \cdot \rangle_M$ is tensor contracted product. For the sake of convenience, we denote $\mathbf{B}_1, \dots, \mathbf{B}_M$ by $\{\mathbf{B}\}$. Fix $\sigma > 0$ and the augmented Lagrangian function of the problem (6) is

$$\begin{aligned} \mathcal{L}(\{\mathbf{B}\}, \boldsymbol{\eta}, \mathcal{W}, \mathbf{g}, \mathcal{G}) = & \mathbb{Q}_{\tau}(\boldsymbol{\eta}) + \sum_{j_1, \dots, j_M} P_{\lambda}(|\mathcal{W}_{j_1, \dots, j_M}|) - \langle \mathbf{g}, \boldsymbol{\eta} - \tilde{\mathbf{y}} + \langle \mathcal{X}_n, [\mathbf{B}_1, \dots, \mathbf{B}_M] \rangle_M \rangle \\ & - \langle \mathcal{G}, \mathcal{W} - [\mathbf{B}_1, \dots, \mathbf{B}_M] \rangle + \frac{\sigma}{2} \|\boldsymbol{\eta} - \tilde{\mathbf{y}} + \langle \mathcal{X}_n, [\mathbf{B}_1, \dots, \mathbf{B}_M] \rangle_M\|_F^2 + \frac{\sigma}{2} \|\mathcal{W} - [\mathbf{B}_1, \dots, \mathbf{B}_M]\|_F^2, \end{aligned}$$

where $\mathbf{g} \in \mathbb{R}^n$ and $\mathcal{G} \in \mathbb{R}^{p_1 \times \dots \times p_M}$ are the Lagrangian multipliers. Through Lemma 1, we let $\mathbf{B}_{-m} = \hat{\mathbf{B}}_M^{(t-1)} \odot \dots \odot \hat{\mathbf{B}}_{m+1}^{(t-1)} \odot \hat{\mathbf{B}}_{m-1}^{(t)} \odot \dots \odot \hat{\mathbf{B}}_1^{(t)} \in \mathbb{R}^{(p_1 \dots p_{m-1} p_{m+1} \dots p_M) \times R}$, $\tilde{\mathbf{x}}_{m,i} = \text{VEC}(\mathcal{X}_{i,(m)} \mathbf{B}_{-m}) \in \mathbb{R}^{p_m R}$, and $\tilde{\mathbf{X}}_m = (\tilde{\mathbf{x}}_{m,1}, \dots, \tilde{\mathbf{x}}_{m,n})^{\top} \in \mathbb{R}^{n \times (p_m R)}$.

All updates for $\{\mathbf{B}\}$, $\boldsymbol{\eta}$, and $\mathcal{W}$ can be derived in closed form. Following Boyd et al. (2011), the iterations for the standard ADMM algorithm are given by

$$\begin{aligned}\{\hat{\mathbf{B}}^{(t)}\} &= \arg \min_{\{\mathbf{B}\}} \mathcal{L}(\{\mathbf{B}\}, \hat{\boldsymbol{\eta}}^{(t-1)}, \hat{\mathcal{W}}^{(t-1)}, \hat{\mathbf{g}}^{(t-1)}, \hat{\mathcal{G}}^{(t-1)}), \\ \hat{\boldsymbol{\eta}}^{(t)} &= \arg \min_{\boldsymbol{\eta}} \mathcal{L}(\{\hat{\mathbf{B}}^{(t)}\}, \boldsymbol{\eta}, \hat{\mathcal{W}}^{(t-1)}, \hat{\mathbf{g}}^{(t-1)}, \hat{\mathcal{G}}^{(t-1)}), \\ \hat{\mathcal{W}}^{(t)} &= \arg \min_{\mathcal{W}} \mathcal{L}(\{\hat{\mathbf{B}}^{(t)}\}, \hat{\boldsymbol{\eta}}^{(t)}, \mathcal{W}, \hat{\mathbf{g}}^{(t-1)}, \hat{\mathcal{G}}^{(t-1)}), \\ \hat{\mathbf{g}}^{(t)} &= \hat{\mathbf{g}}^{(t-1)} - \sigma(\hat{\boldsymbol{\eta}}^{(t)} - \tilde{\mathbf{y}} + \langle \mathcal{X}, \llbracket \hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)} \rrbracket_M \rangle_M), \\ \hat{\mathcal{G}}^{(t)} &= \hat{\mathcal{G}}^{(t-1)} - \sigma(\hat{\mathcal{W}}^{(t)} - \llbracket \hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)} \rrbracket),\end{aligned}$$

where the superscript (t) denotes the t -th iteration. For the update of the parameter set $\{\mathbf{B}\}$, we employ a mode-wise updating strategy. Specifically, for each mode $m = 1, \dots, M$, we consider

$$\begin{aligned}\text{VEC}(\hat{\mathbf{B}}_m) &= \arg \min_{\text{VEC}(\mathbf{B}_m)} \mathcal{L}(\text{VEC}(\mathbf{B}_m), \hat{\boldsymbol{\eta}}^{(t-1)}, \hat{\mathcal{W}}^{(t-1)}, \hat{\mathbf{g}}^{(t-1)}, \hat{\mathcal{G}}^{(t-1)}) \\ &= \arg \min_{\text{VEC}(\mathbf{B}_m)} [\text{VEC}(\hat{\mathcal{G}}_{(m)}^{(t-1)} \mathbf{B}_{-m})^\top - \hat{\mathbf{g}}^{(t-1)\top} \tilde{\mathbf{X}}_m] \text{VEC}(\mathbf{B}_m) + \frac{\sigma}{2} \|\hat{\boldsymbol{\eta}}^{(t-1)} - \tilde{\mathbf{y}} + \tilde{\mathbf{X}}_m \text{VEC}(\mathbf{B}_m)\|_F^2 \\ &\quad + \frac{\sigma}{2} \|\text{VEC}(\hat{\mathcal{W}}_{(m)}^{(t-1)}) - (\mathbf{B}_{-m} \otimes \mathbf{I}_{p_m}) \text{VEC}(\mathbf{B}_m)\|_F^2.\end{aligned}$$

Taking the derivative with respect to $\text{VEC}(\mathbf{B}_m)$ and equating it to zero yields

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \text{VEC}(\mathbf{B}_m)} &= \text{VEC}(\hat{\mathcal{G}}_{(m)}^{(t-1)} \mathbf{B}_{-m}) - \tilde{\mathbf{X}}_m^\top \hat{\mathbf{g}}^{(t-1)} + \sigma \tilde{\mathbf{X}}_m^\top [\hat{\boldsymbol{\eta}}^{(t-1)} - \tilde{\mathbf{y}} + \tilde{\mathbf{X}}_m \text{VEC}(\mathbf{B}_m)] \\ &\quad - \sigma (\mathbf{B}_{-m} \otimes \mathbf{I}_{p_m})^\top [\text{VEC}(\hat{\mathcal{W}}_{(m)}^{(t-1)}) - (\mathbf{B}_{-m} \otimes \mathbf{I}_{p_m}) \text{VEC}(\mathbf{B}_m)] = 0,\end{aligned}$$

and we obtain the closed-form update

$$\begin{aligned}\text{VEC}(\hat{\mathbf{B}}_m^{(t)}) &= [\sigma \tilde{\mathbf{X}}_m^\top \tilde{\mathbf{X}}_m + \sigma (\mathbf{B}_{-m} \otimes \mathbf{I}_{p_m})^\top (\mathbf{B}_{-m} \otimes \mathbf{I}_{p_m})]^{-1} \cdot [\tilde{\mathbf{X}}_m^\top (\hat{\mathbf{g}}^{(t-1)} + \sigma \tilde{\mathbf{y}} - \sigma \hat{\boldsymbol{\eta}}^{(t-1)}) \\ &\quad - \text{VEC}(\hat{\mathcal{G}}_{(m)}^{(t-1)} \mathbf{B}_{-m}) + \sigma (\mathbf{B}_{-m} \otimes \mathbf{I}_{p_m})^\top \text{VEC}(\hat{\mathcal{W}}_{(m)}^{(t-1)})].\end{aligned}$$

For the update of $\boldsymbol{\eta}$, we employ the proximal operator of the check loss function (Gu et al., 2018), as stated in the following lemma.

Lemma 3. Given $\tau \in (0, 1)$ and $\alpha > 0$, the proximal operator of the check loss function is defined as

$$\text{prox}_{\rho_{\tau, \alpha}}(x) = \arg \min_u \rho_{\tau}(u) + \frac{\alpha}{2}(u - x)^2 = \begin{cases} x - \frac{\tau}{\alpha}, & x > \frac{\tau}{\alpha}, \\ 0, & \frac{\tau-1}{\alpha} \leq x \leq \frac{\tau}{\alpha}, \\ x - \frac{\tau-1}{\alpha}, & x < \frac{\tau-1}{\alpha}. \end{cases}$$

The update for $\boldsymbol{\eta}$ can be rewritten as

$$\begin{aligned}\hat{\boldsymbol{\eta}} &= \arg \min_{\boldsymbol{\eta}} \mathcal{L}(\{\hat{\mathbf{B}}^{(t)}\}, \boldsymbol{\eta}, \hat{\mathcal{W}}^{(t-1)}, \hat{\mathbf{g}}^{(t-1)}, \hat{\mathcal{G}}^{(t-1)}) \\ &= \arg \min_{\boldsymbol{\eta}} n\mathbb{Q}_{\tau}(\boldsymbol{\eta}) + \frac{n\sigma}{2} \left\| \boldsymbol{\eta} - \tilde{\mathbf{y}} + \langle \mathcal{X}, \llbracket \hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)} \rrbracket_M \rangle_M - \sigma^{-1} \hat{\mathbf{g}}^{(t-1)} \right\|_2^2.\end{aligned}$$

Applying the proximal operator in Lemma 3, we obtain the update

$$\hat{\boldsymbol{\eta}}^{(t)} = \left(\text{prox}_{\rho_{\tau, n\sigma}} \left(\tilde{\mathbf{y}}_i - \left\langle \llbracket \hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)} \rrbracket, \mathcal{X}_i \right\rangle + \sigma^{-1} \mathbf{g}_i^{(t-1)} \right) \right)_{1 \leq i \leq n}.$$

For the update of $\mathcal{W}$, we employ a simplified version of the proximal operator for nonconvex penalty functions (Wang et al., 2019), as stated in the following lemma.

Lemma 4. *The proximal operator of the SCAD penalty function is given by*

$$\text{prox}_p(x) = \arg \min_u P_\lambda(u) + \frac{\alpha}{2}(u - x)^2 = \begin{cases} \text{sign}(x)(|x| - \lambda)_+, & |x| \leq 2\lambda, \\ \frac{(a-1)x - \text{sign}(x)a\lambda}{a-2}, & 2\lambda < |x| \leq a\lambda, \\ x, & a\lambda < |x|. \end{cases}$$

For the MCP penalty function, the proximal operator is

$$\text{prox}_p(x) = \arg \min_u P_\lambda(u) + \frac{\alpha}{2}(u - x)^2 = \begin{cases} \text{sign}(x)[\frac{a}{a-1}|x| - \frac{a\lambda}{a-1}]_+, & |x| \leq a\lambda, \\ x, & |x| > a\lambda. \end{cases}$$

We can rewrite the update for $\mathcal{W}$ as

$$\begin{aligned} \hat{\mathcal{W}} &= \arg \min_{\mathcal{W}} \mathcal{L}(\{\hat{\mathbf{B}}^{(t)}\}, \hat{\boldsymbol{\eta}}^{(t)}, \mathcal{W}, \hat{\boldsymbol{\xi}}^{(t-1)}, \hat{\mathcal{G}}^{(t-1)}) \\ &= \arg \min_{\mathcal{W}} \sum_{j_1, \dots, j_M} P_\lambda(|\mathcal{W}_{j_1, \dots, j_M}|) + \frac{\sigma}{2} \left\| \mathcal{W} - [\hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)}] - \sigma^{-1} \hat{\mathcal{G}}^{(t-1)} \right\|_2^2. \end{aligned}$$

Applying the proximal operator in Lemma 4, the update for $\mathcal{W}$ is obtained as

$$\hat{\mathcal{W}}^{(t)} = \left(\text{prox}_p \left([\hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)}]_{j_1, \dots, j_M} + \sigma^{-1} \hat{\mathcal{G}}_{j_1, \dots, j_M}^{(t-1)} \right) \right)_{1 \leq j_1 \leq p_1, \dots, 1 \leq j_M \leq p_M}.$$

Secondly, the spline coefficient is updated by

$$\hat{\boldsymbol{\xi}}^{(t)} = \arg \min_{\boldsymbol{\xi}} n^{-1} \sum_{i=1}^n \rho_\tau(\check{y}_i - \boldsymbol{\Pi}(\mathbf{z}_i)^\top \boldsymbol{\xi}),$$

where $\check{y}_i = y_i - \langle [\hat{\mathbf{B}}_1^{(t)}, \dots, \hat{\mathbf{B}}_M^{(t)}], \mathcal{X}_i \rangle$. For the update of $\boldsymbol{\xi}$, we can directly employ conventional linear quantile regression algorithms for efficient estimation.

Note that our alternating update algorithm consists of two key steps: (1) updating the coefficient tensor, and (2) updating the spline coefficients. Through empirical observation, we found that increasing the number of iterations for each alternating step leads to more efficient estimation. Therefore, we implement five iterations for each update step. For the convergence criterion, we adopt an objective-function-based stopping rule:

$$\Delta L = [L(\boldsymbol{\xi}^{(t)}, \mathcal{B}^{(t)}) - L(\boldsymbol{\xi}^{(t-1)}, \mathcal{B}^{(t-1)})] / |L(\boldsymbol{\xi}^{(t-1)}, \mathcal{B}^{(t-1)})|.$$

The iterative updating process terminates when ΔL falls below a predetermined tolerance. The algorithm for solving the TAQR estimator is summarized in Algorithm 1.

For the penalized TAQR estimator, we need to select two parameters: the CP rank R and the tuning parameter λ . Following Li and Zhang (Li and Zhang, 2021) and Liu et al. (Liu et al., 2024), we adopt the Bayesian information criterion,

$$\text{BIC}(R, \lambda) = \log \left[\sum_{i=1}^n \rho_\tau(y_i - \boldsymbol{\Pi}(\mathbf{z}_i)^\top \hat{\boldsymbol{\xi}} - \langle \hat{\mathbf{B}}, \mathcal{X}_i \rangle) \right] + \text{df} \cdot \frac{\log(n)}{2n}, \quad (7)$$

where df represents the degrees of freedom, defined as the total number of nonzero elements in $\hat{\mathbf{B}}$.

3. Numerical Simulation

3.1. Simulation Protocol

In this section, we conduct the numerical simulation to evaluate the finite-sample performance of the proposed TAQR. Specifically, we focus on the capability of TAQR in terms of estimation performance, region selection

Algorithm 1: Algorithm for solving TAQR estimator

Input: dataset $\{y_i, z_i, \mathcal{X}_i\}_{i=1}^n$, rank R , tolerance tol , max iteration T , penalty parameter λ

initialization $\xi^{(0)}, \mathcal{B}^{(0)} = [\hat{\mathcal{B}}_1^{(0)}, \dots, \hat{\mathcal{B}}_M^{(0)}], t \leftarrow 0$;

B-spline transform $\Pi(z_i) = [1, \pi(z_{1,i})^\top, \dots, \pi(z_{p_0,i})^\top]^\top$ for $i = 1, \dots, n$;

while $t < T$ and $\Delta L < tol$ **do**

$\tilde{y}_i = y_i - \Pi(z_i) \hat{\xi}^{(t-1)}$ for $i = 1, \dots, n$;

initialization $\eta^{(0)}, \mathcal{W}^{(0)}, \mathbf{g}^{(0)}, \hat{\mathcal{G}}^{(0)}$;

repeat

for $m \leftarrow 1 - M$ **do**

$\mathcal{B}_{-m} \leftarrow \hat{\mathcal{B}}_M^{(t-1)} \odot \dots \odot \hat{\mathcal{B}}_{m+1}^{(t-1)} \odot \hat{\mathcal{B}}_{m-1}^{(t)} \odot \dots \odot \hat{\mathcal{B}}_1^{(t)}$;

$\tilde{\mathcal{X}}_m \leftarrow [\text{VEC}(\mathcal{X}_{1,[m]} \mathcal{B}_{-m}), \dots, \text{VEC}(\mathcal{X}_{n,[m]} \mathcal{B}_{-m})]^\top$;

$\text{VEC}(\hat{\mathcal{B}}_m^{(t)}) \leftarrow$

$[\sigma \tilde{\mathcal{X}}_m^\top \tilde{\mathcal{X}}_m + \sigma(\mathcal{B}_{-m} \otimes \mathbf{I}_{p_m})^\top (\mathcal{B}_{-m} \otimes \mathbf{I}_{p_m})]^{-1} \cdot [\tilde{\mathcal{X}}_m^\top (\hat{\mathbf{g}}^{(t-1)} + \sigma \tilde{\mathbf{y}} - \sigma \hat{\eta}^{(t-1)}) - \text{VEC}(\hat{\mathcal{G}}_{(m)}^{(t-1)} \mathcal{B}_{-m}) + \sigma(\mathcal{B}_{-m} \otimes \mathbf{I}_{p_m})^\top \text{VEC}(\hat{\mathcal{W}}_{(m)}^{(t-1)})]$;

$\hat{\eta}^{(t)} \leftarrow \text{prox}_{\rho_\tau, n\sigma}(\tilde{y}_i - \langle [\hat{\mathcal{B}}_1^{(t)}, \dots, \hat{\mathcal{B}}_M^{(t)}], \mathcal{X}_i \rangle + \mathbf{g}_i^{(t-1)} / \sigma)_{1 \leq i \leq n}$;

$\hat{\mathcal{W}}^{(t)} \leftarrow \text{prox}_p([\hat{\mathcal{B}}_1^{(t)}, \dots, \hat{\mathcal{B}}_M^{(t)}]_{j_1, \dots, j_M} + \hat{\mathcal{G}}_{j_1, \dots, j_M}^{(t-1)} / \sigma)_{1 \leq j_1 \leq p_1, \dots, 1 \leq j_M \leq p_M}$;

$\hat{\mathbf{g}}^{(t)} \leftarrow \hat{\mathbf{g}}^{(t-1)} - \sigma(\hat{\eta}^{(t)} - \tilde{\mathbf{y}} + \langle \mathcal{X}, [\hat{\mathcal{B}}_1^{(t)}, \dots, \hat{\mathcal{B}}_M^{(t)}] \rangle_M)$;

$\hat{\mathcal{G}}^{(t)} \leftarrow \hat{\mathcal{G}}^{(t-1)} - \sigma(\hat{\mathcal{W}}^{(t)} - [\hat{\mathcal{B}}_1^{(t)}, \dots, \hat{\mathcal{B}}_M^{(t)}])$;

until five iterations;

$\hat{\mathcal{B}}^{(t)} = [\hat{\mathcal{B}}_1^{(t)}, \dots, \hat{\mathcal{B}}_M^{(t)}]$;

repeat

$\tilde{y}_i = y_i - \langle \hat{\mathcal{B}}^{(t)}, \mathcal{X}_i \rangle$ for $i = 1, \dots, n$;

$\hat{\xi}^{(t)} = \arg \min_{\xi} n^{-1} \sum_{i=1}^n \rho_\tau(\tilde{y}_i - \Pi(z_i)^\top \xi)$;

until five iterations;

performance, prediction performance, confidence interval performance, and computational time performance. We generate simulation data from

$$y_i = g_1(z_1) + g_2(z_2) + g_3(z_3) + \langle \mathcal{B}, \mathcal{X}_i \rangle + \varepsilon_i,$$

where $g_1(z_1) = 2(3z_1 - 1.5)^2$, $g_2(z_2) = \sin(2\pi z_2)$, $g_3(z_3) = \cos(2\pi z_3)$, and $z_1, z_2, z_3 \sim U(0, 1)$. We considered three true coefficient tensors of size 32×32 (Case1, Case2, Case3), and the visual representations of these true coefficient tensors are shown in the first column of Figure 2. The entries of the tensor-valued covariate $\mathcal{X}_i$ follow a standard Normal distribution. We consider three distributions of the random error ε_i :

1. $N(0, 1)$: the standard Normal distribution;
2. $\text{Laplace}(0, 2)$: the Laplace distribution with a location parameter of 0 and a scale parameter of 2;
3. $t(3)$: the Student t -distribution with 3 degrees of freedom.

For the training sample size, we consider $n = \{500, 1000, 1500\}$ for the numerical simulation. Prediction performance is evaluated on an independent testing set of 200 samples. We consider three different quantile level $\tau = \{0.2, 0.5, 0.8\}$ in this study. In numerical implementation, we fix the spline basis by employing cubic B-spline basis functions with the number of knots equal to 8, constructed via the `SplineTransformer` function from the `scikit-learn` library, to ensure stable finite-sample performance. While a data-driven selection of k_n would of course be ideal, we adopted an ad-hoc value for the sake of simplicity, and our results remain satisfactory. We acknowledge that performance would likely improve with an optimally selected k_n , and we leave this to future work.

To ensure more reliable comparisons, we replicated the simulation procedure 30 times. We compare the performance of the proposed TAQR with four other regression method: partially linear additive quantile regression model with SCAD penalty (QR-SCAD), partially linear additive quantile regression model with MCP penalty (QR-MCP), element-wise penalized linear quantile tensor regression model with SCAD penalty (QTR-SCAD), and element-wise penalized linear quantile tensor regression model with MCP penalty (QTR-MCP). Due to the space limitation, the majority of simulation results are provided in the Supplementary Material.

3.2. Estimation Performance

We use the root mean squared error $\text{RMSE} = \sqrt{(\prod_{m=1}^M p_m)^{-1} \|\hat{\mathbf{B}} - \mathbf{B}^*\|_F^2}$ to evaluate the estimation performance of the coefficient tensor, and the integrated squared error $\text{ISE} = n_{\text{grid}}^{-1} \sum_{i=1}^{n_{\text{grid}}} \sum_{j=1}^3 \|\hat{g}_j(z_i) - g_j^*(z_i)\|^2$ to evaluate the estimation performance of the nonlinear additive functions, where $n_{\text{grid}} = 1000$, and $z_1, \dots, z_{n_{\text{grid}}}$ are equally spaced points on $[0,1]$. Table 1 reports the estimation performance of the methods under various simulation settings, while Figure 1 show the additive functions estimated by TAQR. Across nearly all simulation scenarios, our proposed TAQR method produced smaller RMSE and ISE values compared to competing methods. This indicates that our method provided more accurate estimates of both the coefficient tensor and the additive functions. The performance gap favoring TAQR consistently widened as the sample size increased, demonstrating its efficiency in leveraging more data. Notably, the relative advantage for our method are strongest when the errors followed heavier-tailed distributions, Laplace(0, 2) and $t(3)$, suggesting that the proposed TAQR method exhibits robust statistical properties. Visually, the curves estimated by TAQR closely tracked the true underlying nonlinear relationships. This demonstrates that our method successfully captured the true additive functions, effectively modeling the flexible nonlinear effects of the vector-valued covariates.

3.3. Region Selection Performance

We use the true positive rate $\text{TPR} = \text{TP}/(\text{TP} + \text{FN})$ and the false positive rate $\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$ to evaluate the region selection performance, where $\text{TP} = \sum_{j_1, \dots, j_M} I(\mathbf{B}_{j_1, \dots, j_M}^* \neq 0) I(\hat{\mathbf{B}}_{j_1, \dots, j_M} \neq 0)$, $\text{FN} = \sum_{j_1, \dots, j_M} I(\mathbf{B}_{j_1, \dots, j_M}^* \neq 0) I(\hat{\mathbf{B}}_{j_1, \dots, j_M} = 0)$, $\text{FP} = \sum_{j_1, \dots, j_M} I(\mathbf{B}_{j_1, \dots, j_M}^* = 0) I(\hat{\mathbf{B}}_{j_1, \dots, j_M} \neq 0)$, and $\text{TN} = \sum_{j_1, \dots, j_M} I(\mathbf{B}_{j_1, \dots, j_M}^* = 0) I(\hat{\mathbf{B}}_{j_1, \dots, j_M} = 0)$. Table 2 summarize the region selection performance, and Figure 2 displays the true and estimated tensor coefficients. TAQR consistently demonstrated superior region selection capabilities. It achieved higher TPR values than QR and QTR, meaning it was better at correctly identifying important regions. Simultaneously, our method maintained lower FPR values, indicating fewer spurious selections. This improvement in precision also became more noticeable with larger sample sizes. Similar to the estimation performance, the region selection performance for TAQR was most evident under Laplace(0, 2) and $t(3)$, reinforcing its robustness in challenging settings. These visualizations clearly illustrate that TAQR recovered the true underlying tensor structure more correctly than QTR and QR. The estimated coefficient tensors from TAQR showed a closer resemblance to the true coefficient tensors, effectively capturing the complex relational dependencies encoded within the tensor covariate, aligning with its quantitative superiority in estimation accuracy.

3.4. Prediction Performance

We use the mean absolute prediction error $\text{MAPE} = \sum_{i=1}^{200} |y_i - \hat{y}_i|/200$ to evaluate the prediction performance. The prediction performance of the model is shown in Figure 3. These MAPE boxplot collectively show that our method generally yielded better prediction performance than QR and QTR. Across different quantiles, sample sizes, and error distributions, predictions based on the TAQR tended to be closer to the true conditional quantiles, as reflected in lower prediction errors. This consistent predictive advantage further validates the practical utility of the TAQR framework.

3.5. Confidence Interval Performance

We use the average lower bound (ALB), average upper bound (AUB), average length (AL), and coverage rate (CR) to evaluate the performance of the confidence interval conducted in Section 2.5. Table 3 reports the performance of the 95% confidence interval across different simulation settings based on 30 replications. Overall, the CR match relatively well to the preassigned level. In particular, under the heavy-tailed error distributions Laplace(0, 2) and $t(3)$, the CR in most settings is close to 0.95, indicating that the conducted confidence interval are robust to heavy-tailed errors. As the sample size increases from 500 to 1500, the AL decreases across all simulation settings, indicating that the confidence intervals become shorter with more data. At the same time, ALB and AUB stay closer to the target value, suggesting that the confidence intervals become more stable. Interestingly, the AL generally becomes larger from Case 1 to Cases 2 and 3, implying that estimation is more challenging in the latter two settings. Nevertheless, the CR remains at a satisfactory level in most simulation settings, which suggests that the conducted confidence intervals continue to perform well under more complex scenarios.

3.6. Computational Time Performance

To compare the computational time of the proposed TAQR method with the traditional vector-valued quantile regression, we report the average computational times (in seconds) across different simulation settings in Table 4. The results clearly show that TAQR consistently outperforms QR. This indicates that even when applying an element-wise penalty to the coefficient tensor, we can still achieve excellent computational efficiency. The main reason for this advantage is that we update the coefficient tensor at the mode-wise level, while the element-wise operation is only applied when penalizing the coefficient tensor.

4. Application to Integrated Genome Study

Complex diseases, including cancer, typically involve genetic abnormalities spanning multiple genomes. The rapid accumulation of diverse omics data has led to the widespread attention in tensor-based integrated genomic analysis (Fang, 2019; Chang et al., 2021; Chen et al., 2025). In this section, we evaluate the practical utility of the proposed TAQR method through an integrated genomic study of lung squamous cell carcinoma (LUSC). LUSC represents a predominant histological subtype of non-small cell lung cancer, characterized by its predilection for central bronchial locations. LUSC demonstrates a strong etiological association with tobacco smoking, with studies indicating that the majority of cases occur in current or former smokers (Kenfield et al., 2008). Therefore, the issue we are interested in is to explore the impact of smoking history and multi-omics biomarkers on the disease-free survival period (DFS) of LUSC.

The multi-omics and clinical data analyzed in this study are obtained from The Cancer Genome Atlas (TCGA, <https://portal.gdc.cancer.gov/>), a landmark collaborative project initiated by the US National Cancer Institute and National Human Genome Research Institute. TCGA systematically collected multi-dimensional omics data from over 20,000 patients across 33 cancer types. This comprehensive dataset incorporates not only molecular profiling data from tumor tissues but also integrates clinical outcome measures and treatment response information, serving as an invaluable resource for cancer biomarker discovery and precision medicine research. To acquire directly analyzable data, we downloaded the Lung Squamous Cell Carcinoma dataset (TCGA, Firehose Legacy) from cBioPortal (<https://www.cbioportal.org/datasets>). After removing samples with missing values, we retained a final cohort of 232 cases for analysis. Smoking history was quantified using pack-year measurements. As shown in Figure 4, descriptive analysis of the response variable DFS revealed a right-skewed distribution with limited outliers, confirming the appropriateness of applying our robust TAQR method to this dataset. For tensor-valued covariate, we integrated multi-omics data from three platforms, including copy number alterations (denoted as CNA), DNA methylation (denoted as Methy), and messenger RNA gene expression (denoted as mRNA). To optimize computational efficiency, we performed feature selection based on variance, retaining the top 200 most variable genes for subsequent analysis. Thus, the final tensor-valued covariate was constructed as a second-order tensor with dimensions 3×200 . Normalization is conducted for the dataset. We construct the following TAQR model:

$$Q_{\tau}(y_i|z_i, \mathcal{X}_i) = g(z_i) + \langle \mathcal{B}, \mathcal{X}_i \rangle,$$

where y_i denotes DFS, z_i denotes the smoking status, and $\mathcal{X}_i$ represents integrated multi-omics data, for $i = 1, \dots, 232$. The model simultaneously estimates the nonlinear additive function $g(\cdot)$ capturing smoking history effects, and the sparse coefficient tensor $\mathcal{B}$ quantifying multi-omics biomarker impacts.

The gene selection results of the model across different quantile levels are presented in Table 5. The selection frequency for each gene was calculated as the number of times it was selected across all quantile levels, $\tau = \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$, divided by the total number of quantiles. Only genes with selection frequencies exceeding 0.5 are displayed in Table 5. Additionally, the specific genes selected by the models across the three quantile levels $\tau = 0.2$, $\tau = 0.5$, and $\tau = 0.8$ are reported in Table 6.

The results in Table 5 reveal several interesting findings. Gene *ARID3C*, a member of the *ARID3* subfamily, is of particular interest, as other *ARID* family members (e.g., *ARID1A*) have been well-documented as tumor suppressors in various cancers (Lin et al., 2014). The loss of these genes has been associated with genomic instability and aberrant cell proliferation. Thus, our findings suggest that further investigation into the relationship between *ARID3C* and *LUSC* may be warranted. Additionally, gene *C9orf131* was selected with a high frequency in our study. Previous research employing statistical analyses has also identified an association between gene *C9orf131* and LUSC prognosis, supporting the biological relevance of our results (Yang et al., 2025). Table 6 further suggests clear heterogeneity in variable selection across quantiles. A small group of genes is repeatedly selected at all three quantile levels, representing

stable biomarkers associated with the conditional distribution of DFS. In contrast, several genes appear only at specific quantiles, indicating that their effects are quantile-dependent. This pattern supports the advantage of quantile regression in uncovering heterogeneous biomarker signals across different parts of the response distribution. Figure 5 and Figure 6 illustrate the nonlinear effects of smoking history on prognosis. Our analysis reveals a non-monotonic relationship between smoking history and DFS, with the nonlinear effects exhibiting heterogeneity across different quantile levels.

Although TAQR-SCAD and TAQR-MCP yield slightly different selected biomarkers, the discrepancies mainly occur for genes with moderate or weak effects. This is not surprising because SCAD and MCP have different shrinkage profiles and thresholding behaviors, which may treat coefficients near the selection boundary differently in finite samples. Importantly, the two penalties identify a substantial overlap of key biomarkers, indicating that the main biological conclusions are robust to the choice of nonconvex penalty.

5. Conclusion and Future Work

We have proposed the novel tensor regression model, TAQR, for modeling the conditional quantile of a scalar-valued response given both vector-valued and tensor-valued covariates. Building upon the B-spline approximation and the CP decomposition, we have developed an estimator incorporating a nonconvex penalty and provide a corresponding solution algorithm based on the ADMM. Under high dimension settings, we have established the asymptotic properties of the proposed estimator. Extensive simulations demonstrated its performance under limited sample sizes, and real-world applications on the genome study validated its applicability.

Looking ahead, there are several intriguing research directions:

1. Nonparametric Models: Although we have focused on semi-parametric models, future research could investigate modeling the tensor component under a nonparametric setting.
2. Tensor-Related Norms: TAQR currently employs sparse penalties like SCAD and MCP. Exploring penalty structures that incorporate tensor norms could enhance its performance.
3. Tensor-Valued Response Variables: While our study exclusively examines tensor-valued covariates, exploring scenarios involving tensor-valued response variables holds promise for further research.

Acknowledgments

We thank the Editor, Associate Editor and referees.

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Appendix

Proofs of Theorems

In this section, we present detailed proofs of Theorems 1 and 2. For convenience, we introduce the following notations. Denote by $U = \Pi(\Pi^T \Pi)^{-1} \Pi^T$, $\Pi = [\Pi(z_1), \dots, \Pi(z_n)]^T$, $\tilde{X}^* = (I_n - U)\tilde{X}$, and $\Psi = U\tilde{X} = (\psi_1, \dots, \psi_n)^T$. Let $L^P(\xi, \phi) = \sum_{i=1}^n \rho_\tau(y_i - \Pi(z_i)^T \xi - \tilde{X}_i^T \phi) + n \sum_{j=1}^{p_n} P_\lambda(|\phi_j|)$ and $\tilde{L}^P(\phi) = \sum_{i=1}^n \rho_\tau(y_i - \Pi(z_i)^T \xi^* - \tilde{X}_i^{*T} \phi -$

$\psi_i^\top \phi^*) + n \sum_{j=1}^{p_n} P_\lambda(|\phi_j|)$. Define

$$(\hat{\xi}, \hat{\phi}) = \arg \min_{\xi, \phi} L^P(\xi, \phi), \text{ and } \tilde{\phi} = \arg \min_{\phi} \tilde{L}^P(\phi).$$

Proof of Theorem 1. Let $\alpha_n = (n^{-1/2} + a_n)(R \sum_{m=1}^M p_m + k_n)^{1/2}$, $\mathbf{u} = \alpha_n^{-1}(\xi - \xi^*)$, and $\mathbf{v} = \alpha_n^{-1}(\phi - \phi^*)$. Consider the ball centered at zero: $R(\mathbf{u}, \mathbf{v}) = \{(\mathbf{u}, \mathbf{v}) : \|\mathbf{u}\|_2 + \|\mathbf{v}\|_2 = C\}$. We will prove that for any $\mu > 0$, there exists a large constant C , such that

$$\Pr \left\{ \inf_{(\mathbf{u}, \mathbf{v}) \in R(\mathbf{u}, \mathbf{v})} L^P(\xi^* + \alpha_n \mathbf{u}, \phi^* + \alpha_n \mathbf{v}) - L^P(\xi^*, \phi^*) > 0 \right\} \geq 1 - \mu. \quad (8)$$

That is, with probability tending to one, there exists a local minimum $(\hat{\xi}, \hat{\phi})$ in the ball $\{(\xi^* + \alpha_n \mathbf{u}, \phi^* + \alpha_n \mathbf{v}) : \|\mathbf{u}\|_2 + \|\mathbf{v}\|_2 \leq C\}$ such that $\|\hat{\xi} - \xi^*\|_2 + \|\hat{\phi} - \phi^*\|_2 = O_P(\alpha_n)$.

Let

$$D(\mathbf{u}, \mathbf{v}) = L^P(\xi^* + \alpha_n \mathbf{u}, \phi^* + \alpha_n \mathbf{v}) - L^P(\xi^*, \phi^*) = S(\mathbf{u}, \mathbf{v}) + R(\mathbf{v}), \quad (9)$$

where $S(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \rho_\tau(y_i - \Pi(\mathbf{z}_i)^\top (\xi^* + \alpha_n \mathbf{u}) - \tilde{\mathbf{X}}_i^\top (\phi^* + \alpha_n \mathbf{v})) - \sum_{i=1}^n \rho_\tau(y_i - \Pi(\mathbf{z}_i)^\top \xi^* - \tilde{\mathbf{X}}_i^\top \phi^*)$, and $R(\mathbf{v}) = n \sum_{j=1}^{p_n} [P_\lambda(|\phi_j^* + \alpha_n \mathbf{v}_j|) - P_\lambda(|\phi_j^*|)]$. Applying the Knight identity (Knight, 1998),

$$\rho_\tau(u - v) - \rho_\tau(u) = v(I(u < 0) - \tau) + \int_0^v [I(u \leq t) - I(u \leq 0)] dt,$$

we obtain that

$$S(\mathbf{u}, \mathbf{v}) \equiv I_1 + I_2 + I_3,$$

where $I_1 = \alpha_n \sum_{i=1}^n \Pi(\mathbf{z}_i)^\top \mathbf{u} (I(\varepsilon_i < 0) - \tau)$, $I_2 = \alpha_n \sum_{i=1}^n \tilde{\mathbf{X}}_i^\top \mathbf{v} (I(\varepsilon_i < 0) - \tau)$, and $I_3 = \sum_{i=1}^n \int_0^{\alpha_n \Pi(\mathbf{z}_i)^\top \mathbf{u} + \alpha_n \tilde{\mathbf{X}}_i^\top \mathbf{v}} [I(\varepsilon_i \leq t) - I(\varepsilon_i \leq 0)] dt$. In the following we evaluate each I_k for $k = 1, 2, 3$.

Let $d_{1,i} = \Pi(\mathbf{z}_i)^\top \mathbf{u}$. Then $|d_{1,i}| \leq \|\Pi(\mathbf{z}_i)\|_2 \|\mathbf{u}\|_2 = O_P(\|\mathbf{u}\|_2)$. It follows that $E(I_1) = 0$ and there exists constant c_1 such that

$$E(I_1^2) \leq c_1 \alpha_n^2 \sum_{i=1}^n E[\Pi(\mathbf{z}_i)^\top \mathbf{u}]^2 = c_1 \alpha_n^2 \mathbf{u}^\top \sum_{i=1}^n E[\Pi(\mathbf{z}_i) \Pi(\mathbf{z}_i)^\top] \mathbf{u} = o_P(n \alpha_n^2 \|\mathbf{u}\|_2^2).$$

Next, we consider the second term I_2 . Let $d_{2,i} = \tilde{\mathbf{X}}_i^\top \mathbf{v}$. Then $|d_{2,i}| \leq \|\tilde{\mathbf{X}}_i\|_2 \|\mathbf{v}\|_2 = O_P(\|\mathbf{v}\|_2)$. It follows that $E(I_2) = 0$ and there exists some constant c_2 such that

$$E(I_2^2) \leq c_2 \alpha_n^2 \sum_{i=1}^n E(\tilde{\mathbf{X}}_i^\top \mathbf{v})^2 = c_2 \alpha_n^2 \mathbf{v}^\top \left(\sum_{i=1}^n \tilde{\mathbf{X}}_i \tilde{\mathbf{X}}_i^\top \right) \mathbf{v} = o_P(n \alpha_n^2 \|\mathbf{v}\|_2^2).$$

For the third term I_3 , by the Taylor expansion, we establish that

$$\begin{aligned} E(I_3) &= E \sum_{i=1}^n \int_0^{\alpha_n \Pi(\mathbf{z}_i)^\top \mathbf{u} + \alpha_n \tilde{\mathbf{X}}_i^\top \mathbf{v}} [I(\varepsilon_i \leq t) - I(\varepsilon_i \leq 0)] dt \\ &= \sum_{i=1}^n \int_0^{\alpha_n \Pi(\mathbf{z}_i)^\top \mathbf{u} + \alpha_n \tilde{\mathbf{X}}_i^\top \mathbf{v}} [F_\varepsilon(t | \mathbf{z}_i, \mathcal{X}_i) - F_\varepsilon(0 | \mathbf{z}_i, \mathcal{X}_i)] dt \\ &= \sum_{i=1}^n \int_0^{\alpha_n \Pi(\mathbf{z}_i)^\top \mathbf{u} + \alpha_n \tilde{\mathbf{X}}_i^\top \mathbf{v}} f_\varepsilon(0 | \mathbf{z}_i, \mathcal{X}_i) t (1 + o_P(1)) dt \\ &= \frac{1}{2} \sum_{i=1}^n f_\varepsilon(0 | \mathbf{z}_i, \mathcal{X}_i) \alpha_n^2 [\mathbf{u}^\top \Pi(\mathbf{z}_i) \Pi(\mathbf{z}_i)^\top \mathbf{u} + \mathbf{v}^\top \tilde{\mathbf{X}}_i \tilde{\mathbf{X}}_i^\top \mathbf{v} + 2(\Pi(\mathbf{z}_i)^\top \mathbf{u})(\tilde{\mathbf{X}}_i^\top \mathbf{v})] (1 + o_P(1)). \end{aligned}$$

It follows that

$$S(\mathbf{u}, \mathbf{v}) = \frac{1}{2} \sum_{i=1}^n f_{\varepsilon}(0|z_i, \mathcal{X}_i) \alpha_n^2 [\mathbf{u}^{\top} \mathbf{\Pi}(z_i) \mathbf{\Pi}(z_i)^{\top} \mathbf{u} + \mathbf{v}^{\top} \tilde{\mathbf{X}}_i \tilde{\mathbf{X}}_i^{\top} \mathbf{v} + 2(\mathbf{\Pi}(z_i)^{\top} \mathbf{u})(\tilde{\mathbf{X}}_i^{\top} \mathbf{v})] (1 + o_P(1)) \\ + o_P(n\alpha_n^2)(\|\mathbf{u}\|_2 + \|\mathbf{v}\|_2). \quad (10)$$

By $P_{\lambda}(0) = 0$ and the Taylor expansion, it is easy to see that

$$R(\mathbf{v}) \geq \sum_{j=1}^{q_n} \left[n\alpha_n P'_{\lambda}(|\phi_j^*|) \mathbf{v}_j \text{sgn}(\phi_j^*) + \frac{1}{2} n\alpha_n^2 P''_{\lambda}(|\phi_j^*|) \mathbf{v}_j^2 (1 + o(1)) \right] \\ \geq -[n\alpha_n^2 \|\mathbf{v}\|_2 + o_P(n\alpha_n^2)].$$

In summary, when $\|\mathbf{u}\|_2$ and $\|\mathbf{v}\|_2$ are sufficiently large, $S(\mathbf{u}, \mathbf{v})$ is guaranteed to dominate $R(\mathbf{v})$. Therefore, we can select a sufficiently large C to ensure that (8) holds. $\square$

Lemma 5. Suppose Conditions 1 - 7 hold. If $\lambda \rightarrow 0$ and $\lambda[n(R \sum_{m=1}^M p_m)^{-1}]^{1/2} \rightarrow \infty$ as $n \rightarrow \infty$, then for any given ξ and ϕ satisfying $\|\xi - \xi^*\|_2 + \|\phi - \phi^*\|_2 = O_P(\alpha_n)$ and for any constant C , we have

$$L^P(\xi, (\phi_{\mathbb{A}}^{\top}, \mathbf{0}^{\top})^{\top}) = \min_{\|\phi_{\mathbb{A}^c}\| \leq C\alpha_n} L^P(\xi, (\phi_{\mathbb{A}}^{\top}, \phi_{\mathbb{A}^c}^{\top})^{\top}).$$

Proof of Lemma 5. Let $\mathbf{v}_{\mathbb{A}} = \alpha_n^{-1}(\phi_{\mathbb{A}} - \phi_{\mathbb{A}}^*)$, $\mathbf{v}_{\mathbb{A}^c} = \alpha_n^{-1}(\phi_{\mathbb{A}^c} - \phi_{\mathbb{A}^c}^*)$, and $\mathbf{v} = (\mathbf{v}_{\mathbb{A}}^{\top}, \mathbf{v}_{\mathbb{A}^c}^{\top})^{\top}$. By the definition of $L^P(\xi, \phi)$, we have

$$L^P(\xi, (\phi_{\mathbb{A}}^{\top}, \mathbf{0}^{\top})^{\top}) - L^P(\xi, (\phi_{\mathbb{A}}^{\top}, \phi_{\mathbb{A}^c}^{\top})^{\top}) = \Pi_1 - \Pi_2 - \Pi_3,$$

where

$$\Pi_1 = \sum_{i=1}^n \rho_{\tau} [y_i - \mathbf{\Pi}(z_i)^{\top} (\xi^* + \alpha_n \mathbf{u}) - \tilde{\mathbf{X}}_i^{\top} ((\phi_{\mathbb{A}}^* + \alpha_n \mathbf{v}_{\mathbb{A}})^{\top}, \mathbf{0}^{\top})] - \sum_{i=1}^n \rho_{\tau} [y_i - \mathbf{\Pi}(z_i)^{\top} \xi^* - \tilde{\mathbf{X}}_i^{\top} (\phi_{\mathbb{A}}^{*\top}, \mathbf{0}^{\top})],$$

$$\Pi_2 = \sum_{i=1}^n \rho_{\tau} [y_i - \mathbf{\Pi}(z_i)^{\top} (\xi^* + \alpha_n \mathbf{u}) - \tilde{\mathbf{X}}_i^{\top} ((\phi_{\mathbb{A}}^* + \alpha_n \mathbf{v}_{\mathbb{A}})^{\top}, (\phi_{\mathbb{A}^c}^* + \alpha_n \mathbf{v}_{\mathbb{A}^c})^{\top})] - \sum_{i=1}^n \rho_{\tau} [y_i - \mathbf{\Pi}(z_i)^{\top} \xi^* - \tilde{\mathbf{X}}_i^{\top} (\phi_{\mathbb{A}}^{*\top}, \phi_{\mathbb{A}^c}^{*\top})],$$

and $\Pi_3 = n \sum_{j=q_n+1}^{p_n} P_{\lambda}(|\phi_j|)$.

From eq.(10), we obtain that

$$\Pi_2 = \frac{1}{2} \sum_{i=1}^n f_{\varepsilon}(0|z_i, \mathcal{X}_i) \alpha_n^2 (\mathbf{u}^{\top} \mathbf{\Pi}(z_i) \mathbf{\Pi}(z_i)^{\top} \mathbf{u} + \mathbf{v}^{\top} \tilde{\mathbf{X}}_i \tilde{\mathbf{X}}_i^{\top} \mathbf{v} + 2(\mathbf{\Pi}(z_i)^{\top} \mathbf{u})(\tilde{\mathbf{X}}_i^{\top} \mathbf{v})) (1 + o_P(1)) + o_P(n\alpha_n^2)(\|\mathbf{u}\|_2 + \|\mathbf{v}\|_2).$$

Since $\|\mathbf{u}\|_2 + \|\mathbf{v}\|_2 = O_P(1)$, we get $\Pi_1 = O_P(n\alpha_n^2)$ and $\Pi_2 = O_P(n\alpha_n^2)$. For the third term, we have

$$\Pi_3 = n \sum_{j=q_n+1}^{p_n} P'_{\lambda}(|\phi_j^{\natural}|) |\phi_j^{\natural}| \\ \geq p_n^{3/2} \left(R \sum_{m=1}^M p_m \right)^{1/2} (np_n^{-3})^{1/2} \left(\frac{n}{R \sum_{m=1}^M p_m} \right)^{1/2} \lambda \left(\liminf_{n \rightarrow +\infty} \liminf_{\beta \rightarrow 0+} \frac{P'_{\lambda}(\beta)}{\lambda} \right) \cdot \left(\sum_{j=q_n+1}^{p_n} |\phi_j^{\natural}| \right) (1 + o(1)),$$

where $0 < \phi_j^{\natural} < |\phi_j|$, $j = q_n + 1, \dots, p_n$. Note that $\lambda n^{1/2} (R \sum_{m=1}^M p_m)^{-1/2} \rightarrow \infty$ and $p_n^3/n \rightarrow 0$. It follows that $p_n^{3/2} (R \sum_{m=1}^M p_m)^{1/2} (np_n^{-3})^{1/2} (n(R \sum_{m=1}^M p_m)^{-1})^{1/2} \lambda$ is of higher order than $O_P(n\alpha_n^2)$. Then $L^P(\xi, (\phi_{\mathbb{A}}^{\top}, \mathbf{0}^{\top})^{\top}) - L^P(\xi, (\phi_{\mathbb{A}}^{\top}, \phi_{\mathbb{A}^c}^{\top})^{\top})$ is dominated by the negative term Π_3 . The proof of Lemma 5 is completed. $\square$

Lemma 6. *If Condition 1 - 7 holds, then*

$$\sqrt{n}e_n' \mathbf{K}^{-1/2}(\mathbf{K} + \mathbf{\Omega}_\lambda)[\tilde{\boldsymbol{\phi}}_\mathbb{A} - \boldsymbol{\phi}_\mathbb{A}^* + (\mathbf{K} + \mathbf{\Omega}_\lambda)^{-1} \mathbf{c}_n] \rightarrow N(\mathbf{0}, \tau(1 - \tau)),$$

where $\mathbf{K}$ is defined in Theorem 2.

Proof of Lemma 6. Let $\mathbf{v} = \alpha_n^{-1}(\boldsymbol{\phi} - \boldsymbol{\phi}^*)$. Partition the vectors $\mathbf{v} = (\mathbf{v}_\mathbb{A}^\top, \mathbf{v}_{\mathbb{A}^c}^\top)^\top$. Similar to (9), we define

$$\tilde{D}(\mathbf{v}) = \tilde{S}(\mathbf{v}) + R(\mathbf{v}),$$

where

$$\tilde{S}(\mathbf{v}) = \sum_{i=1}^n \rho_\tau \left(y_i - \mathbf{\Pi}(\mathbf{z}_i)^\top \boldsymbol{\xi}^* - \tilde{\mathbf{X}}_i^\top \boldsymbol{\phi}^* - \alpha_n \tilde{\mathbf{X}}_i^{*\top} \mathbf{v} \right) - \sum_{i=1}^n \rho_\tau \left(y_i - \mathbf{\Pi}(\mathbf{z}_i)^\top \boldsymbol{\xi}^* - \tilde{\mathbf{X}}_i^\top \boldsymbol{\phi}^* \right).$$

By $P_\lambda(0) = 0$, we can write

$$\tilde{D}((\mathbf{v}_\mathbb{A}^\top, \mathbf{0}^\top)^\top) = \tilde{S}((\mathbf{v}_\mathbb{A}^\top, \mathbf{0}^\top)^\top) + R(\mathbf{v}_\mathbb{A}),$$

where $R(\mathbf{v}_\mathbb{A}) = n \sum_{j=1}^{q_n} [P_\lambda(|\boldsymbol{\phi}_j^* + \alpha_n \mathbf{v}_j|) - P_\lambda(|\boldsymbol{\phi}_j^*|)]$. By Conditions 5 and 7, and by taking Taylor's expansion for $R(\mathbf{v}_\mathbb{A})$ at $\mathbf{v}_\mathbb{A} = 0$, we obtain that

$$R(\mathbf{v}_\mathbb{A}) = n\alpha_n \mathbf{c}_n^\top \mathbf{v}_\mathbb{A} + \frac{1}{2} n\alpha_n^2 \mathbf{v}_\mathbb{A}^\top \mathbf{\Omega}_\lambda \mathbf{v}_\mathbb{A} (1 + o(1)).$$

Then the minimizer $\tilde{\mathbf{v}} = (\tilde{\mathbf{v}}_\mathbb{A}^\top, \mathbf{0}^\top)^\top$ of $\tilde{D}(\mathbf{v})$ satisfies the score equations $\nabla \tilde{D}((\tilde{\mathbf{v}}_\mathbb{A}^\top, \mathbf{0}^\top)^\top) = 0$. That is,

$$\frac{1}{n} \sum_{i=1}^n \tilde{\mathbf{X}}_{\mathbb{A}i}^* [\tau - I(\varepsilon_i + R(\mathbf{z}_i) < \alpha_n \tilde{\mathbf{X}}_{\mathbb{A}i}^{*\top} \tilde{\mathbf{v}}_\mathbb{A})] = \mathbf{c}_n + \alpha_n \mathbf{\Omega}_\lambda \tilde{\mathbf{v}}_\mathbb{A} (1 + o_P(1)), \quad (11)$$

where $R(\mathbf{z}_i) = g^*(\mathbf{z}_i) - \mathbf{\Pi}(\mathbf{z}_i)^\top \boldsymbol{\xi}^*$. The left-hand side of (11) can be rewritten as

$$\frac{1}{n} \sum_{i=1}^n \tilde{\mathbf{X}}_{\mathbb{A}i}^* [\tau - I(\varepsilon_i + R(\mathbf{z}_i) < \alpha_n \tilde{\mathbf{X}}_{\mathbb{A}i}^{*\top} \tilde{\mathbf{v}}_\mathbb{A})] = -n^{-1/2} \mathbf{z}_n + \mathbf{B}_{n1} + \mathbf{B}_{n2} + o_P(\alpha_n), \quad (12)$$

where

$$\begin{aligned} \mathbf{z}_n &= n^{-1/2} \sum_{i=1}^n \tilde{\mathbf{X}}_{\mathbb{A}i}^* [I(\varepsilon_i + R(\mathbf{z}_i) < 0) - \tau], \\ \mathbf{B}_{n1} &= -\alpha_n n^{-1} \sum_{i=1}^n f(0|\mathbf{z}_i, \mathcal{X}_i) \tilde{\mathbf{X}}_{\mathbb{A}i}^* \tilde{\mathbf{X}}_{\mathbb{A}i}^{*\top} \tilde{\mathbf{v}}_\mathbb{A}, \\ \mathbf{B}_{n2} &= n^{-1} \sum_{i=1}^n f(0|\mathbf{z}_i, \mathcal{X}_i) \tilde{\mathbf{X}}_{\mathbb{A}i}^* R(\mathbf{z}_i). \end{aligned}$$

According to Lemma 3 in Sherwood and Wang (2016a), we have

$$n^{-1} \tilde{\mathbf{X}}_\mathbb{A}^{*\top} \mathbf{D}_n \tilde{\mathbf{X}}_\mathbb{A}^* = n^{-1} \mathbf{\Delta}_n^\top \mathbf{D}_n \mathbf{\Delta}_n + o_P(1). \quad (13)$$

Meanwhile, by direct calculation of the mean and variance similar to the proof of Theorem 2 in Jiang et al. (2012), we can get $\mathbf{B}_{n2} = o_P(\alpha_n)$. Considering the asymptotic variance of the over-parameterized structure in tensor regression, refer to Theorem 1 in Lu et al. (2020), we establish that

$$e_n^\top \mathbf{K}^{-1/2} \mathbf{z}_n \rightarrow N(0, \tau(1 - \tau)). \quad (14)$$

where $\mathbf{e}_n$ is a unit vector. Combining (11), (12), (13), and (14) leads to

$$\alpha_n(\mathbf{K} + \mathbf{\Omega}_\lambda)\tilde{\mathbf{v}}_\mathbb{A} + \mathbf{c}_n = -n^{-1/2}\mathbf{z}_n + o_P(n^{-1/2}).$$

Note that $\tilde{\mathbf{v}}_\mathbb{A} = \alpha_n^{-1}(\tilde{\boldsymbol{\phi}}_\mathbb{A} - \boldsymbol{\phi}_\mathbb{A}^*)$. It follows that

$$\sqrt{n}\mathbf{e}_n^\top \mathbf{K}^{-1/2}(\mathbf{K} + \mathbf{\Omega}_\lambda)[\tilde{\boldsymbol{\phi}}_\mathbb{A} - \boldsymbol{\phi}_\mathbb{A}^* + (\mathbf{K} + \mathbf{\Omega}_\lambda)^{-1}\mathbf{c}_n] \rightarrow N(\mathbf{0}, \tau(1 - \tau)).$$

□

Proof of Theorem 2. It follows from Lemma 5 that Theorem 2(i) holds. By Lemma S4, we obtain the asymptotic normality of $\tilde{\boldsymbol{\phi}}_\mathbb{A}$. Thus, it remains to prove that $\|\tilde{\boldsymbol{\phi}}_\mathbb{A} - \boldsymbol{\phi}_\mathbb{A}^*\| = o_P(1)$. Let $\boldsymbol{\theta}_1 = \sqrt{n}(\boldsymbol{\phi}_\mathbb{A} - \boldsymbol{\phi}_\mathbb{A}^*)$. Then

$$\sum_{i=1}^n \rho_\tau \left(y_i - \mathbf{\Pi}(\mathbf{z}_i)^\top \boldsymbol{\xi} - \tilde{\mathbf{X}}_{\mathbb{A}i}^\top \boldsymbol{\phi}_\mathbb{A} \right) = \sum_{i=1}^n \rho_\tau(\varepsilon_i - n^{-1/2} \tilde{\mathbf{X}}_{\mathbb{A}i}^\top \boldsymbol{\theta}_1 - \tilde{\mathbf{\Pi}}(\mathbf{z}_i)^\top \boldsymbol{\theta}_2 + R(\mathbf{z}_i)),$$

where $\tilde{\mathbf{\Pi}}(\mathbf{z}_i) = (\mathbf{\Pi}^\top \mathbf{\Pi})^{-1} \mathbf{\Pi}(\mathbf{z}_i)$, and $\boldsymbol{\theta}_2 = \mathbf{\Pi}^\top \mathbf{\Pi}(\boldsymbol{\xi} - \boldsymbol{\xi}^*) + \mathbf{\Pi}^\top \tilde{\mathbf{X}}_\mathbb{A}(\boldsymbol{\phi}_\mathbb{A} - \boldsymbol{\phi}_\mathbb{A}^*)$. Define

$$\tilde{Q}_i(\boldsymbol{\theta}_1, \tilde{\boldsymbol{\theta}}_1, \boldsymbol{\theta}_2) = \rho_\tau(\varepsilon_i - n^{-1/2} \tilde{\mathbf{X}}_{\mathbb{A}i}^\top \boldsymbol{\theta}_1 - \tilde{\mathbf{\Pi}}(\mathbf{z}_i)^\top \boldsymbol{\theta}_2 + R(\mathbf{z}_i)) - \rho_\tau(\varepsilon_i - n^{-1/2} \tilde{\mathbf{X}}_{\mathbb{A}i}^\top \tilde{\boldsymbol{\theta}}_1 - \tilde{\mathbf{\Pi}}(\mathbf{z}_i)^\top \boldsymbol{\theta}_2 + R(\mathbf{z}_i)).$$

It suffices to show that, for any positive constants M and C ,

$$\Pr \left\{ \inf_{\|\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1\| \geq M, \|\boldsymbol{\theta}_2\| \leq C\sqrt{n}\alpha_n} \sum_{i=1}^n \tilde{Q}_i(\boldsymbol{\theta}_1, \tilde{\boldsymbol{\theta}}_1, \boldsymbol{\theta}_2) > 0 \right\} \rightarrow 1. \quad (15)$$

By Lemma B.3 and Lemma B.4 in Sherwood and Wang (2016b), we have

$$\begin{aligned} \sup_{\|\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1\| \leq M, \|\boldsymbol{\theta}_2\| \leq C\sqrt{n}\alpha_n} \left| \sum_{i=1}^n \left[\tilde{Q}_i(\boldsymbol{\theta}_1, \tilde{\boldsymbol{\theta}}_1, \boldsymbol{\theta}_2) + n^{-1/2} \tilde{\mathbf{X}}_{\mathbb{A}i}^\top (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1)(\tau - I(\varepsilon_i < 0)) \right] \right. \\ \left. - \frac{1}{2} \left(\boldsymbol{\theta}_1^\top \mathbf{G}_n \boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1^\top \mathbf{G}_n \tilde{\boldsymbol{\theta}}_1 \right) (1 + o_P(1)) \right| = o_P(1). \end{aligned} \quad (16)$$

Refer to Lemma 5 in Sherwood and Wang (2016a), we notice that

$$n^{-1/2}(\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1)^\top \sum_{i=1}^n \tilde{\mathbf{X}}_{\mathbb{A}i}^* \{\tau - I(\varepsilon_i < 0)\} = (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1)^\top \mathbf{G}_n \tilde{\boldsymbol{\theta}}_1 (1 + o_P(1)). \quad (17)$$

Combining (16) and (17) leads to

$$\sup_{\|\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1\| \leq M, \|\boldsymbol{\theta}_2\| \leq C\sqrt{n}\alpha_n} \left| \sum_{i=1}^n \tilde{Q}_i(\boldsymbol{\theta}_1, \tilde{\boldsymbol{\theta}}_1, \boldsymbol{\theta}_2) + (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1)^\top \mathbf{G}_n \tilde{\boldsymbol{\theta}}_1 (1 + o_P(1)) - \frac{1}{2} \left(\boldsymbol{\theta}_1^\top \mathbf{G}_n \boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1^\top \mathbf{G}_n \tilde{\boldsymbol{\theta}}_1 \right) \right| = o_P(1),$$

which implies that

$$\sup_{\|\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1\| \leq M, \|\boldsymbol{\theta}_2\| \leq C\sqrt{n}\alpha_n} \left| \sum_{i=1}^n \tilde{Q}_i(\boldsymbol{\theta}_1, \tilde{\boldsymbol{\theta}}_1, \boldsymbol{\theta}_2) - \frac{1}{2} (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1)^\top \mathbf{G}_n (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1) (1 + o_P(1)) \right| = o_P(1).$$

By Condition 2, for any $\|\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1\| \geq M$ and some positive constant C , we have

$$\frac{1}{2} (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1)^\top \mathbf{G}_n (\boldsymbol{\theta}_1 - \tilde{\boldsymbol{\theta}}_1) > CM.$$

Therefore, (15) hold. □

Supplementary

Supplementary material has to be uploaded separately: some additional simulation results are included in the supplementary materials.

Table 1: The RMSE and ISE results under different simulation settings. The random error is following $t(3)$. The values are the averages of 30 repetitions, with standard deviations given in parentheses.

B	n	method	$\tau = 0.2$		$\tau = 0.5$		$\tau = 0.8$	
			RMSE	ISE	RMSE	ISE	RMSE	ISE
Case1	500	QR-SCAD	0.073 (0.023)	1.017 (0.787)	0.053 (0.009)	0.535 (0.223)	0.074 (0.018)	0.944 (0.734)
		QR-MCP	0.072 (0.010)	0.688 (0.287)	0.062 (0.011)	0.521 (0.237)	0.079 (0.020)	0.836 (0.430)
		QTR-SCAD	0.018 (0.005)	6.274 (1.872)	0.020 (0.010)	6.062 (1.853)	0.029 (0.012)	5.992 (2.284)
		QTR-MCP	0.017 (0.005)	5.897 (1.590)	0.020 (0.013)	5.501 (1.163)	0.023 (0.007)	6.025 (1.575)
		TAQR-SCAD	0.009 (0.002)	0.503 (0.394)	0.008 (0.002)	0.794 (1.092)	0.010 (0.002)	0.724 (0.515)
		TAQR-MCP	0.010 (0.002)	0.681 (0.668)	0.008 (0.002)	0.604 (0.414)	0.010 (0.003)	0.619 (0.448)
	1000	QR-SCAD	0.053 (0.011)	0.522 (0.454)	0.038 (0.009)	0.393 (0.375)	0.054 (0.011)	0.695 (0.711)
		QR-MCP	0.055 (0.010)	0.627 (0.641)	0.039 (0.007)	0.431 (0.273)	0.053 (0.010)	0.506 (0.380)
		QTR-SCAD	0.011 (0.003)	5.998 (1.775)	0.014 (0.005)	6.230 (1.905)	0.016 (0.005)	5.576 (1.516)
		QTR-MCP	0.011 (0.003)	5.990 (1.671)	0.015 (0.008)	6.611 (1.705)	0.019 (0.009)	5.934 (1.958)
		TAQR-SCAD	0.006 (0.001)	0.255 (0.272)	0.005 (0.001)	0.272 (0.210)	0.006 (0.001)	0.274 (0.273)
		TAQR-MCP	0.007 (0.001)	0.298 (0.346)	0.006 (0.001)	0.253 (0.188)	0.007 (0.001)	0.276 (0.214)
	1500	QR-SCAD	0.028 (0.006)	0.183 (0.119)	0.014 (0.002)	0.270 (0.160)	0.028 (0.004)	0.306 (0.372)
		QR-MCP	0.025 (0.005)	0.423 (0.588)	0.014 (0.002)	0.359 (0.689)	0.026 (0.005)	0.281 (0.202)
		QTR-SCAD	0.009 (0.003)	6.543 (2.548)	0.010 (0.003)	5.642 (1.517)	0.014 (0.006)	5.898 (1.459)
		QTR-MCP	0.010 (0.003)	5.595 (1.479)	0.010 (0.003)	5.718 (1.571)	0.016 (0.006)	5.692 (1.414)
		TAQR-SCAD	0.005 (0.001)	0.342 (0.394)	0.004 (0.001)	0.261 (0.264)	0.005 (0.001)	0.226 (0.196)
		TAQR-MCP	0.006 (0.001)	0.178 (0.248)	0.005 (0.001)	0.147 (0.097)	0.006 (0.001)	0.248 (0.206)
Case2	500	QR-SCAD	0.061 (0.012)	0.650 (0.331)	0.052 (0.012)	0.436 (0.247)	0.062 (0.019)	0.850 (1.179)
		QR-MCP	0.064 (0.011)	0.672 (0.359)	0.061 (0.022)	0.668 (0.515)	0.069 (0.015)	0.749 (0.571)
		QTR-SCAD	0.029 (0.008)	6.266 (2.662)	0.035 (0.013)	5.363 (1.162)	0.055 (0.023)	6.556 (1.756)
		QTR-MCP	0.031 (0.012)	6.418 (2.048)	0.042 (0.018)	5.746 (1.494)	0.063 (0.030)	5.893 (1.654)
		TAQR-SCAD	0.014 (0.002)	0.638 (0.586)	0.012 (0.002)	0.641 (0.654)	0.013 (0.002)	0.744 (0.713)
		TAQR-MCP	0.014 (0.003)	0.653 (0.653)	0.012 (0.002)	0.858 (0.825)	0.014 (0.003)	0.562 (0.553)
	1000	QR-SCAD	0.051 (0.011)	0.448 (0.213)	0.037 (0.007)	0.470 (0.436)	0.048 (0.011)	0.475 (0.439)
		QR-MCP	0.050 (0.013)	0.423 (0.240)	0.037 (0.007)	0.369 (0.266)	0.054 (0.009)	0.565 (0.424)
		QTR-SCAD	0.014 (0.003)	6.402 (1.842)	0.017 (0.007)	7.015 (2.227)	0.026 (0.013)	5.357 (1.395)
		QTR-MCP	0.016 (0.006)	6.061 (2.070)	0.020 (0.010)	5.512 (1.898)	0.027 (0.013)	5.739 (1.890)
		TAQR-SCAD	0.008 (0.001)	0.319 (0.292)	0.007 (0.001)	0.324 (0.331)	0.008 (0.001)	0.261 (0.246)
		TAQR-MCP	0.009 (0.002)	0.370 (0.350)	0.008 (0.001)	0.313 (0.384)	0.008 (0.001)	0.355 (0.478)
	1500	QR-SCAD	0.024 (0.004)	0.365 (0.396)	0.012 (0.002)	0.241 (0.206)	0.025 (0.006)	0.334 (0.231)
		QR-MCP	0.026 (0.006)	0.217 (0.150)	0.012 (0.002)	0.280 (0.256)	0.025 (0.005)	0.303 (0.255)
		QTR-SCAD	0.012 (0.002)	5.910 (1.655)	0.014 (0.006)	5.806 (1.841)	0.019 (0.007)	5.576 (1.729)
		QTR-MCP	0.013 (0.004)	6.248 (2.290)	0.014 (0.006)	5.928 (1.575)	0.022 (0.013)	6.011 (2.124)
		TAQR-SCAD	0.007 (0.001)	0.267 (0.270)	0.005 (0.001)	0.224 (0.183)	0.006 (0.001)	0.196 (0.170)
		TAQR-MCP	0.007 (0.001)	0.307 (0.373)	0.006 (0.001)	0.259 (0.233)	0.007 (0.001)	0.190 (0.284)
Case3	500	QR-SCAD	0.084 (0.016)	0.855 (0.388)	0.065 (0.016)	0.658 (0.577)	0.082 (0.017)	0.924 (0.471)
		QR-MCP	0.100 (0.016)	1.037 (0.465)	0.079 (0.018)	0.621 (0.310)	0.101 (0.023)	1.194 (0.920)
		QTR-SCAD	0.069 (0.016)	5.898 (1.551)	0.071 (0.023)	6.320 (2.074)	0.140 (0.061)	6.441 (2.185)
		QTR-MCP	0.067 (0.017)	6.239 (1.715)	0.079 (0.029)	5.885 (1.349)	0.140 (0.053)	6.058 (1.936)
		TAQR-SCAD	0.024 (0.004)	1.040 (0.957)	0.024 (0.004)	0.818 (0.745)	0.024 (0.004)	1.050 (0.776)
		TAQR-MCP	0.025 (0.003)	1.286 (1.366)	0.026 (0.004)	1.091 (0.704)	0.025 (0.003)	1.940 (4.069)
	1000	QR-SCAD	0.066 (0.013)	0.758 (0.642)	0.045 (0.007)	0.432 (0.365)	0.071 (0.016)	0.644 (0.409)
		QR-MCP	0.065 (0.011)	0.588 (0.347)	0.048 (0.009)	0.459 (0.286)	0.065 (0.015)	0.565 (0.320)
		QTR-SCAD	0.025 (0.007)	5.555 (1.511)	0.026 (0.005)	5.571 (1.741)	0.041 (0.018)	5.803 (1.623)
		QTR-MCP	0.024 (0.006)	6.476 (2.147)	0.030 (0.008)	6.030 (1.725)	0.047 (0.020)	5.781 (1.687)
		TAQR-SCAD	0.012 (0.002)	0.443 (0.534)	0.011 (0.002)	0.294 (0.276)	0.012 (0.002)	0.419 (0.523)
		TAQR-MCP	0.013 (0.002)	0.709 (1.462)	0.012 (0.001)	0.376 (0.374)	0.013 (0.001)	0.407 (0.814)
	1500	QR-SCAD	0.031 (0.005)	0.299 (0.248)	0.018 (0.002)	0.342 (0.411)	0.033 (0.009)	0.266 (0.192)
		QR-MCP	0.032 (0.007)	0.296 (0.287)	0.017 (0.002)	0.315 (0.254)	0.031 (0.006)	0.243 (0.211)
		QTR-SCAD	0.018 (0.004)	6.018 (1.780)	0.019 (0.004)	6.590 (2.594)	0.026 (0.007)	5.526 (1.244)
		QTR-MCP	0.018 (0.004)	5.999 (1.317)	0.024 (0.010)	6.230 (1.890)	0.025 (0.007)	6.040 (1.637)
		TAQR-SCAD	0.009 (0.001)	0.276 (0.225)	0.008 (0.001)	0.203 (0.197)	0.010 (0.002)	0.256 (0.190)
		TAQR-MCP	0.011 (0.001)	0.216 (0.202)	0.009 (0.001)	0.204 (0.162)	0.011 (0.001)	0.205 (0.167)

Table 2: The TPR and FPR results under different simulation settings. The random error is following $t(3)$. The values are the averages of 30 repetitions, with standard deviations given in parentheses.

B	n	method	$\tau = 0.2$		$\tau = 0.5$		$\tau = 0.8$	
			TPR	FPR	TPR	FPR	TPR	FPR
Case1	500	QR-SCAD	0.974 (0.034)	0.136 (0.019)	1.000 (0.000)	0.109 (0.012)	0.976 (0.029)	0.140 (0.019)
		QR-MCP	0.989 (0.017)	0.100 (0.013)	0.999 (0.005)	0.091 (0.010)	0.980 (0.030)	0.101 (0.012)
		QTR-SCAD	1.000 (0.000)	0.001 (0.002)	1.000 (0.000)	0.002 (0.006)	1.000 (0.000)	0.004 (0.008)
		QTR-MCP	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.003 (0.008)	1.000 (0.000)	0.002 (0.005)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
	1000	QR-SCAD	0.972 (0.027)	0.028 (0.007)	0.996 (0.012)	0.012 (0.005)	0.970 (0.028)	0.030 (0.009)
		QR-MCP	0.972 (0.026)	0.029 (0.009)	0.994 (0.011)	0.013 (0.005)	0.975 (0.023)	0.031 (0.008)
		QTR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		QTR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.000 (0.002)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
	1500	QR-SCAD	0.998 (0.007)	0.013 (0.006)	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.013 (0.004)
		QR-MCP	0.999 (0.005)	0.012 (0.006)	1.000 (0.000)	0.000 (0.001)	0.999 (0.005)	0.013 (0.005)
		QTR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		QTR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
Case2	500	QR-SCAD	0.981 (0.033)	0.131 (0.013)	0.999 (0.007)	0.111 (0.014)	0.989 (0.019)	0.134 (0.022)
		QR-MCP	0.994 (0.014)	0.104 (0.012)	0.999 (0.007)	0.096 (0.017)	0.985 (0.020)	0.107 (0.013)
		QTR-SCAD	1.000 (0.000)	0.009 (0.007)	1.000 (0.000)	0.011 (0.012)	1.000 (0.000)	0.026 (0.019)
		QTR-MCP	1.000 (0.000)	0.009 (0.009)	1.000 (0.000)	0.014 (0.013)	1.000 (0.000)	0.029 (0.022)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.001)
	1000	QR-SCAD	0.968 (0.037)	0.030 (0.009)	0.995 (0.012)	0.012 (0.003)	0.971 (0.033)	0.028 (0.007)
		QR-MCP	0.968 (0.037)	0.028 (0.007)	0.998 (0.009)	0.012 (0.006)	0.965 (0.036)	0.031 (0.007)
		QTR-SCAD	1.000 (0.000)	0.001 (0.001)	1.000 (0.000)	0.001 (0.004)	1.000 (0.000)	0.004 (0.007)
		QTR-MCP	1.000 (0.000)	0.001 (0.002)	1.000 (0.000)	0.002 (0.004)	1.000 (0.000)	0.004 (0.006)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
	1500	QR-SCAD	1.000 (0.000)	0.011 (0.004)	1.000 (0.000)	0.000 (0.000)	0.998 (0.009)	0.012 (0.005)
		QR-MCP	0.996 (0.011)	0.012 (0.005)	1.000 (0.000)	0.000 (0.000)	0.999 (0.007)	0.013 (0.004)
		QTR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.001 (0.002)	1.000 (0.000)	0.001 (0.002)
		QTR-MCP	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.003 (0.005)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
Case3	500	QR-SCAD	0.967 (0.021)	0.127 (0.016)	0.998 (0.005)	0.115 (0.018)	0.973 (0.024)	0.132 (0.014)
		QR-MCP	0.968 (0.028)	0.099 (0.013)	1.000 (0.000)	0.088 (0.013)	0.973 (0.027)	0.100 (0.016)
		QTR-SCAD	1.000 (0.000)	0.054 (0.024)	1.000 (0.000)	0.054 (0.032)	0.995 (0.014)	0.159 (0.088)
		QTR-MCP	1.000 (0.000)	0.048 (0.024)	1.000 (0.000)	0.057 (0.038)	0.995 (0.017)	0.146 (0.078)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.001 (0.001)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.001 (0.001)	1.000 (0.000)	0.000 (0.000)
	1000	QR-SCAD	0.968 (0.026)	0.034 (0.009)	0.998 (0.005)	0.015 (0.005)	0.961 (0.032)	0.038 (0.011)
		QR-MCP	0.974 (0.022)	0.033 (0.008)	0.993 (0.011)	0.015 (0.005)	0.973 (0.027)	0.034 (0.010)
		QTR-SCAD	1.000 (0.000)	0.004 (0.005)	1.000 (0.000)	0.004 (0.004)	1.000 (0.000)	0.016 (0.017)
		QTR-MCP	1.000 (0.000)	0.003 (0.004)	1.000 (0.000)	0.005 (0.005)	1.000 (0.000)	0.016 (0.016)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
	1500	QR-SCAD	0.999 (0.003)	0.014 (0.005)	1.000 (0.000)	0.000 (0.001)	0.996 (0.009)	0.013 (0.006)
		QR-MCP	0.998 (0.007)	0.014 (0.005)	1.000 (0.000)	0.000 (0.001)	0.999 (0.005)	0.013 (0.006)
		QTR-SCAD	1.000 (0.000)	0.001 (0.002)	1.000 (0.000)	0.001 (0.001)	1.000 (0.000)	0.003 (0.005)
		QTR-MCP	1.000 (0.000)	0.000 (0.001)	1.000 (0.000)	0.003 (0.006)	1.000 (0.000)	0.002 (0.002)
		TAQR-SCAD	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)
		TAQR-MCP	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)	1.000 (0.000)	0.000 (0.000)

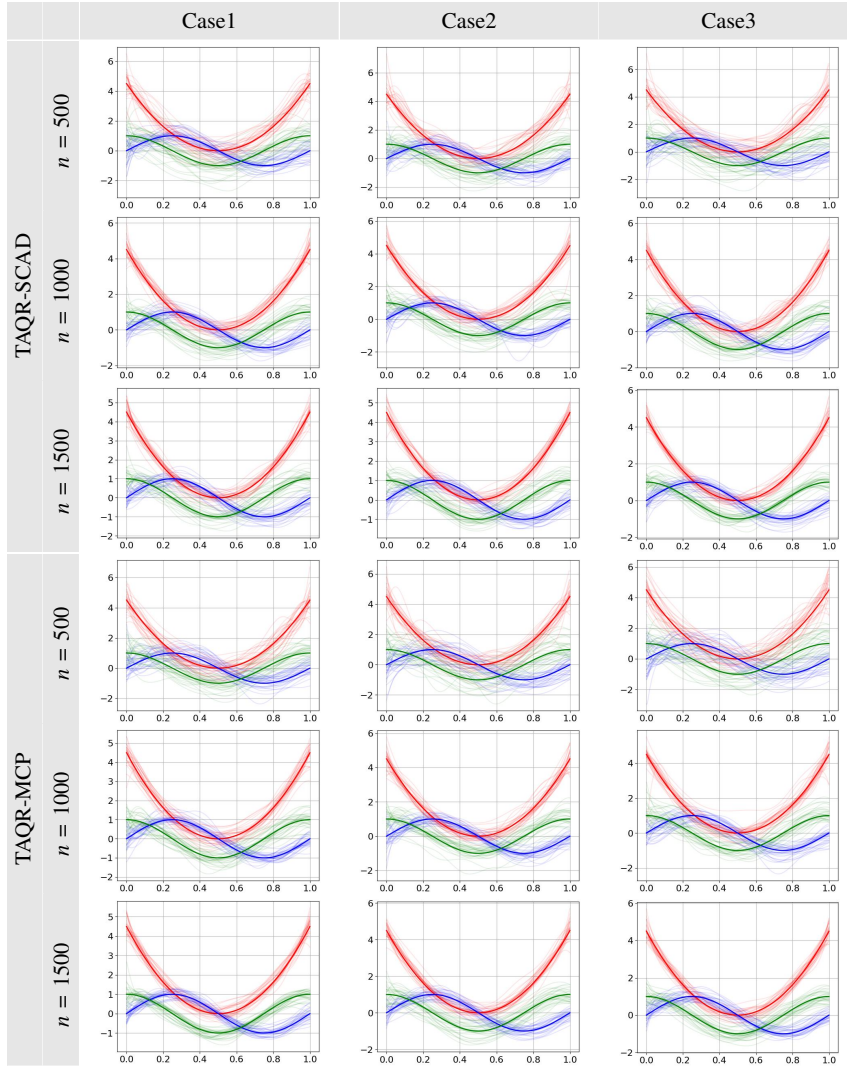


Fig. 1: Estimated additive components under $\tau = 0.5$. Red: $g_1(\cdot)$; blue: $g_2(\cdot)$; green: $g_3(\cdot)$. The random error is following $t(3)$. Bold lines represent the true functions and thin lines represent estimates from 30 replications.

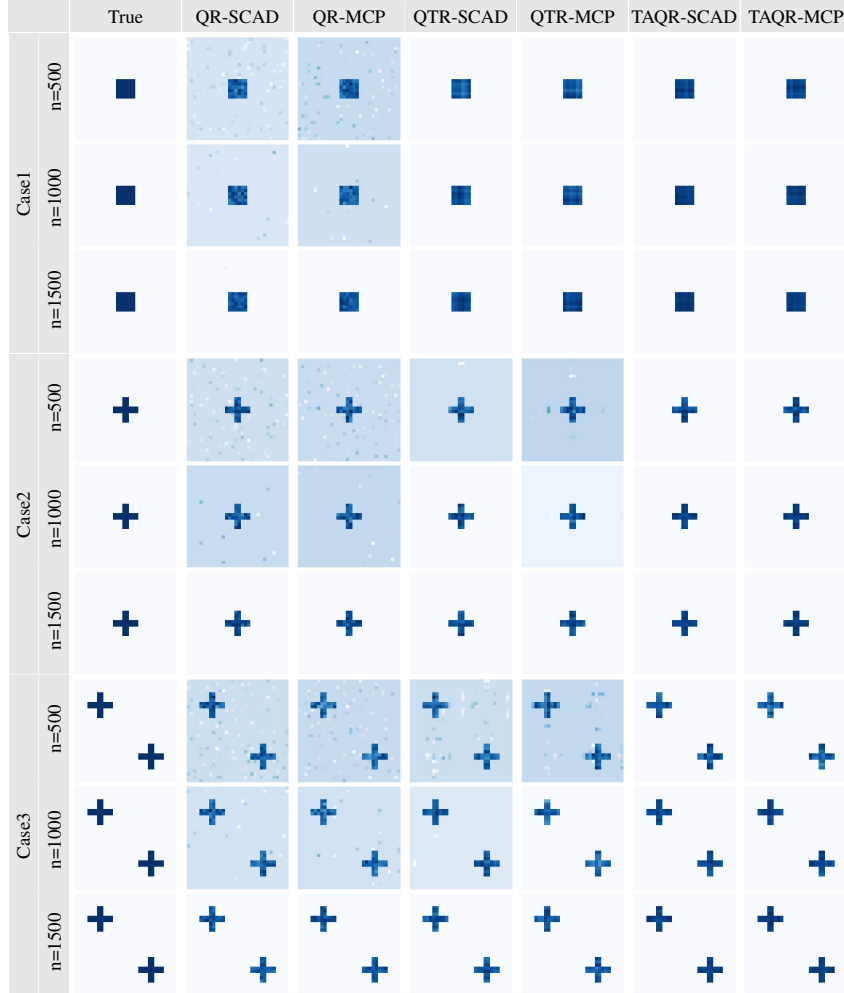


Fig. 2: True coefficient tensors and estimated coefficient tensors. The random error is following $t(3)$ and quantile level $\tau = 0.5$. Each estimate represents the median result of the corresponding RMSE.

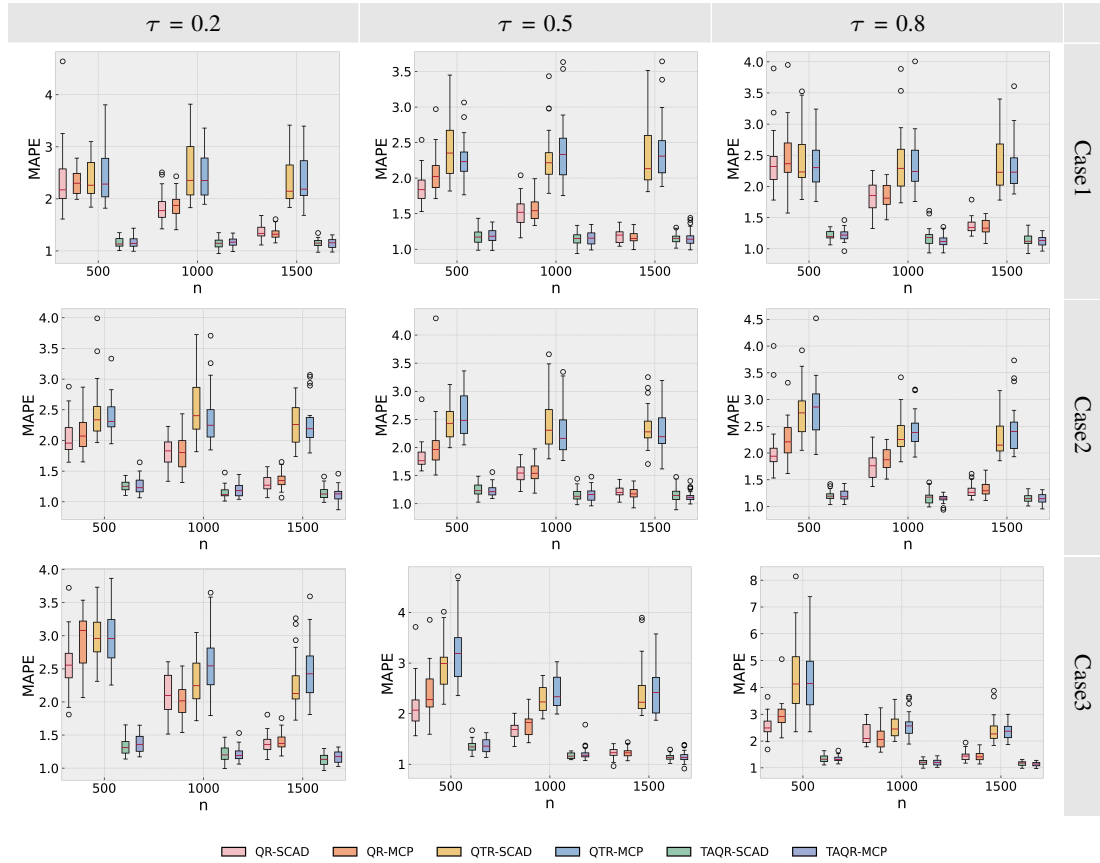


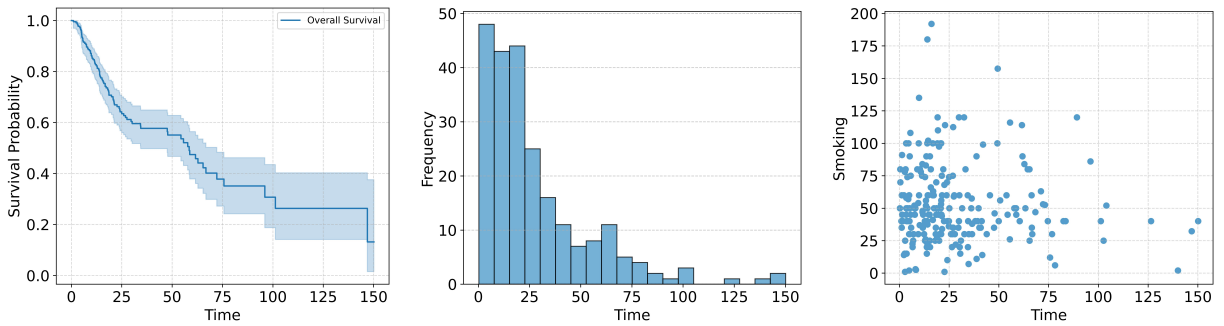
Fig. 3: Boxplot of MAPE under various simulation settings. The random error is following $t(3)$.

Table 3: Confidence interval performance of model under various simulation settings, $\tau = 0.5$.

error	method	n	Case 1				Case 2				Case 3			
			ALB	AUB	AL	CR	ALB	AUB	AL	CR	ALB	AUB	AL	CR
$N(0, 1)$	TAQR-SCAD	500	0.932	1.064	0.131	0.900	0.890	1.118	0.229	0.933	0.882	1.119	0.238	0.933
		1000	0.959	1.053	0.094	0.900	0.922	1.079	0.158	0.967	0.914	1.074	0.160	0.967
		1500	0.967	1.038	0.071	0.933	0.938	1.068	0.130	0.967	0.942	1.074	0.132	0.933
	TAQR-MCP	500	0.925	1.050	0.125	0.800	0.867	1.096	0.230	0.933	0.882	1.128	0.246	0.933
		1000	0.938	1.032	0.094	0.833	0.899	1.058	0.159	0.933	0.905	1.068	0.163	1.000
		1500	0.949	1.019	0.071	0.800	0.911	1.042	0.131	0.967	0.921	1.052	0.131	0.900
Laplace(0, 2)	TAQR-SCAD	500	0.850	1.111	0.261	0.900	0.731	1.239	0.507	0.967	0.629	1.263	0.634	0.967
		1000	0.901	1.087	0.186	0.933	0.836	1.175	0.339	1.000	0.828	1.173	0.345	0.967
		1500	0.927	1.073	0.146	0.967	0.869	1.137	0.269	0.967	0.870	1.140	0.270	0.967
	TAQR-MCP	500	0.823	1.127	0.304	0.967	0.659	1.173	0.514	0.933	0.664	1.309	0.645	0.900
		1000	0.881	1.090	0.209	0.933	0.838	1.174	0.335	1.000	0.808	1.154	0.346	0.967
		1500	0.900	1.058	0.157	0.967	0.875	1.140	0.265	0.967	0.847	1.120	0.273	0.933
$t(3)$	TAQR-SCAD	500	0.932	1.074	0.142	0.933	0.868	1.157	0.290	0.933	0.805	1.137	0.332	0.900
		1000	0.940	1.054	0.114	0.967	0.896	1.087	0.191	1.000	0.929	1.125	0.197	1.000
		1500	0.955	1.038	0.084	0.967	0.920	1.072	0.152	0.933	0.914	1.069	0.154	0.967
	TAQR-MCP	500	0.909	1.072	0.163	0.900	0.839	1.120	0.281	0.900	0.797	1.127	0.330	0.900
		1000	0.931	1.029	0.098	0.933	0.876	1.070	0.194	0.967	0.880	1.078	0.198	0.967
		1500	0.947	1.032	0.085	0.967	0.919	1.073	0.154	0.933	0.904	1.060	0.157	0.967

Table 4: Computational time comparison of model fitting under various simulation settings, $n = 500$.

B	method	$N(0, 1)$			Laplace(0, 2)			$t(3)$		
		$\tau = 0.2$	$\tau = 0.5$	$\tau = 0.8$	$\tau = 0.2$	$\tau = 0.5$	$\tau = 0.8$	$\tau = 0.2$	$\tau = 0.5$	$\tau = 0.8$
Case1	QR-SCAD	10.521	9.893	9.627	9.144	9.097	9.511	9.908	9.441	9.411
	QR-MCP	9.629	9.794	9.366	9.577	9.174	9.433	9.387	9.151	9.216
	TAQR-SCAD	2.956	2.411	2.451	2.314	2.504	2.584	2.472	2.279	2.617
	TAQR-MCP	2.455	2.601	2.649	2.310	2.760	2.335	2.411	2.393	2.871
Case2	QR-SCAD	10.526	10.628	10.633	9.625	9.572	9.748	9.093	9.737	9.317
	QR-MCP	9.632	10.538	9.905	9.365	9.502	9.087	9.446	9.318	9.418
	TAQR-SCAD	3.750	4.184	3.527	4.682	4.686	4.801	4.150	3.616	4.025
	TAQR-MCP	3.995	3.969	4.296	4.752	4.670	4.703	4.212	3.830	3.788
Case3	QR-SCAD	10.675	10.346	10.741	9.303	8.882	9.135	9.283	9.326	9.673
	QR-MCP	10.344	10.028	10.212	9.569	9.149	9.021	9.583	9.247	9.419
	TAQR-SCAD	7.764	7.273	7.724	7.742	7.712	7.671	7.700	7.600	7.709
	TAQR-MCP	7.653	7.451	7.602	7.622	7.635	7.709	7.681	7.671	7.650

**Fig. 4:** Descriptive statistical analysis of the real dataset. Left: The Kaplan–Meier curve illustrates DFS for patients. Middle: Histogram of DFS. Right: Scatter plot of DFS versus pack-years of smoking.**Table 5:** The gene selection results of TAQR-SCAD and TAQR-MCP across different quantiles.

TAQR-SCAD			TAQR-MCP		
gene	platform	frequency	gene	platform	frequency
<i>ARID3C</i>	mRNA	1.00	<i>C9orf131</i>	mRNA	0.89
<i>C9orf131</i>	mRNA	1.00	<i>HOXA13</i>	mRNA	0.89
<i>HOXA13</i>	mRNA	0.89	<i>PCDHA5</i>	mRNA	0.78
<i>RNASE3</i>	mRNA	0.89	<i>RNASE3</i>	mRNA	0.67
<i>SLC22A9</i>	mRNA	0.78	<i>ALMS1P</i>	mRNA	0.67
<i>C9orf131</i>	Methy	0.67	<i>ARID3C</i>	mRNA	0.67
<i>PCDHA5</i>	mRNA	0.56	<i>SLC22A9</i>	mRNA	0.56

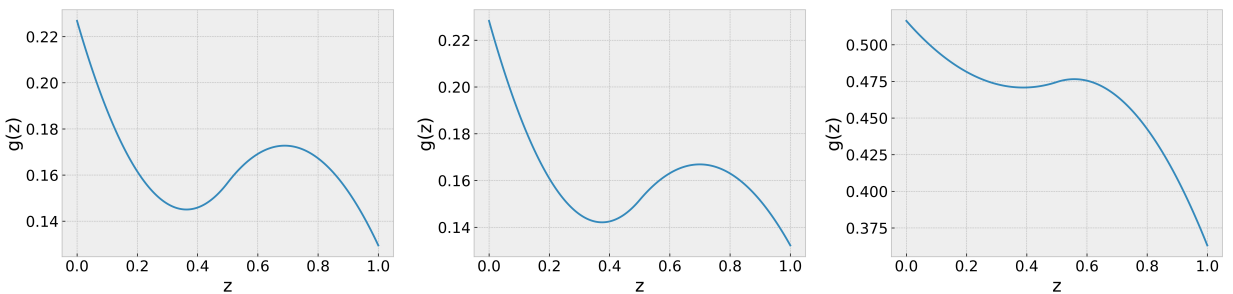
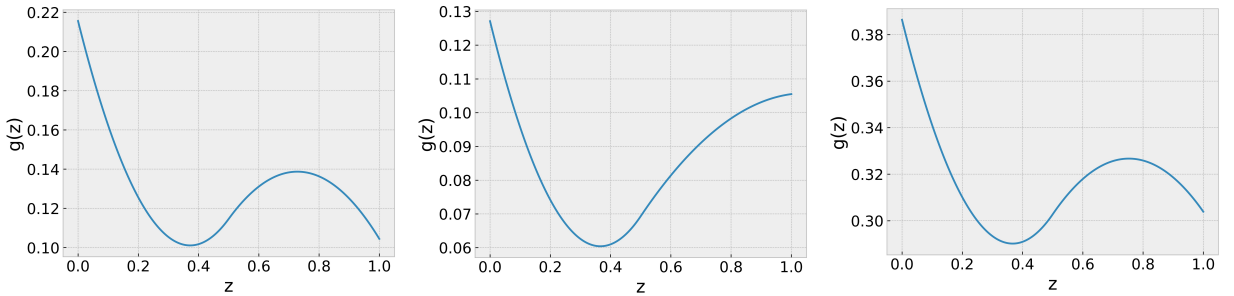
**Fig. 5:** The nonlinear effects estimated by TAQR-SCAD. Left: $\tau = 0.2$; Middle: $\tau = 0.5$; Right: $\tau = 0.8$.

Table 6: Genes selected by TAQR-SCAD and TAQR-MCP at the three quantile levels $\tau = 0.2$, $\tau = 0.5$, and $\tau = 0.8$. Bold indicates genes selected at all three quantile levels.

Method	τ	Gene
TAQR-SCAD	0.2	ARID3C (mRNA) , C9orf131 (mRNA) , HOXA13 (mRNA) , RNASE3 (mRNA) , <i>ALMS1P (mRNA)</i> , <i>ARID3C (Methy)</i> , <i>SLC22A9 (mRNA)</i> , <i>SLC22A9 (Methy)</i> , <i>RLN1 (mRNA)</i> , <i>SLC22A14 (mRNA)</i> , <i>C9orf131 (Methy)</i>
	0.5	ARID3C (mRNA) , C9orf131 (mRNA) , HOXA13 (mRNA) , RNASE3 (mRNA) , <i>POM121L1P (Methy)</i> , <i>ARID3C (Methy)</i> , <i>PCDHA5 (mRNA)</i> , <i>C9orf131 (Methy)</i> , <i>HOXA13 (Methy)</i> , <i>RNASE3 (Methy)</i>
	0.8	ARID3C (mRNA) , C9orf131 (mRNA) , HOXA13 (mRNA) , RNASE3 (mRNA) , <i>ALMS1P (mRNA)</i> , <i>PCDHA5 (mRNA)</i> , <i>SLC22A9 (mRNA)</i> , <i>CDK15 (mRNA)</i> , <i>TTC36 (mRNA)</i> , <i>AP1M2 (mRNA)</i> , <i>MYLPF (mRNA)</i> , <i>NAT2 (mRNA)</i> , <i>KLRC4 (mRNA)</i> , <i>HOXD11 (mRNA)</i>
TAQR-MCP	0.2	C9orf131 (mRNA) , HOXA13 (mRNA) , PCDHA5 (mRNA) , RNASE3 (mRNA) , <i>ALMS1P (mRNA)</i> , <i>MYLPF (Methy)</i> , <i>SLC22A9 (mRNA)</i> , <i>RLN1 (mRNA)</i> , <i>C9orf50 (mRNA)</i> , <i>TICAM2 (mRNA)</i> , <i>ARID3C (Methy)</i> , <i>SLC22A14 (mRNA)</i> , <i>ACSBG2 (mRNA)</i> , <i>GTF2A1L (mRNA)</i>
	0.5	C9orf131 (mRNA) , HOXA13 (mRNA) , PCDHA5 (mRNA) , RNASE3 (mRNA) , <i>POM121L1P (Methy)</i> , <i>ARID3C (mRNA)</i> , <i>ZNF556 (mRNA)</i> , <i>SLC22A14 (mRNA)</i> , <i>C9orf131 (Methy)</i> , <i>ACSBG2 (mRNA)</i> , <i>HOXA13 (Methy)</i> , <i>POM121L1P (mRNA)</i>
	0.8	C9orf131 (mRNA) , HOXA13 (mRNA) , PCDHA5 (mRNA) , RNASE3 (mRNA) , <i>FAM66A (mRNA)</i> , <i>TOMM20L (mRNA)</i> , <i>MYLPF (mRNA)</i> , <i>SLC22A9 (mRNA)</i> , <i>FOXO4L6 (mRNA)</i> , <i>TTC36 (mRNA)</i> , <i>ARID3C (mRNA)</i> , <i>C9orf131 (Methy)</i> , <i>NAT2 (mRNA)</i> , <i>KLRC4 (mRNA)</i> , <i>ALMS1P (mRNA)</i>

**Fig. 6:** The nonlinear effects estimated by TAQR-MCP. Left: $\tau = 0.2$; Middle: $\tau = 0.5$; Right: $\tau = 0.8$.